Bioenergetics of Exercising Humans

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ABSTRACT:

Human muscles, limbs and supporting ventilatory, cardiovascular, and metabolic systems are well adapted for walking, and there is reasonable transfer of efficiency of movement to bicycling. Our efficiency and economy of movement of bipedal walking (≈30%) are far superior to those of apes. This overall body efficiency during walking and bicycling represents the multiplicative interaction of a phosphorylative coupling efficiency of ≈60%, and a mechanical coupling efficiency of ≈50%. These coupling efficiencies compare well with those of other species adapted for locomotion. We are capable runners, but our speed and power are inferior to carnivorous and omnivorous terrestrial mammalian quadrupeds because of biomechanical and physiological constraints. But, because of our metabolic plasticity (i.e., the ability to switch among carbohydrate (CHO)- and lipid-derived energy sources) our endurance capacity is very good by comparison to most mammals, but inferior to highly adapted species such as wolves and migratory birds. Our ancestral ability for hunting and gathering depends on strategy and capabilities in the areas of thermoregulation, and metabolic plasticity. Clearly, our competitive advantage of survival in the biosphere depends in intelligence and behavior. Today, those abilities that served early hunter-gatherers make for interesting athletic competitions due to wide variations in human phenotypes. In contemporary society, the stresses of regular physical exercise serve to minimize morbidities and mortality associated with physical inactivity, overnutrition, and aging. © 2012 American Physiological Society. *Compr Physiol* 2:537-562, 2012.

Introduction

Whether stated or not, the matter of human bioenergetics is implicitly fundamental to studies of human performance and metabolism in health and disease. How the body acquires, stores, and utilizes energy has allowed our ancestors to hunt and gather, to escape predation, to survive famines and migrations, to deal with heat, cold, the hypoxia of high altitude, and to carry children to term. Accordingly, the history of studies of human bioenergetics is broad and deep, ranging from archeology and anthropology, to studies of tissue metabolism and muscle mitochondrial energetics. Related are studies from comparative physiology on the metabolic capabilities of terrestrial, aquatic, and aerial animals. Human history, as reflected in our human phenotypes confers success to some of us in daily life as well as very different types of sports and games; but, ironically, in contemporary developed societies some of the same capabilities that led to biological success of past generations can lead to the development of chronic diseases related to physical inactivity.

In this article emphasis is on the bioenergetics of humans. So far as is known, we share similar cardiovascular and muscle designs as with other animal species, but we are adapted for upright, bipedal locomotion freeing the hands and arms for important tasks. But, how do we compare to other animals in terms of physical prowess? Though celebrated in legend (e.g., Phidippides of Marathon fame) and in the annals of the modern athletics (e.g., Abebe Bikila in the 1960 and 1964 Olympiads) our capacity for locomotion is paltry compared to those of other species that are more powerful, faster (17) and possess greater endurance (106). For example, since the Englishman Sir Roger Bannister achieved the first sub-4 min mile (3:59.4 in 1954), that performance of running 4 min at a speed of 15 mph has been repeated and surpassed countless times. However, at the time of this writing the World Record for the 2-mile run is 7:58.61 by the Kenyan Daniel Komen in 1997. That is an extraordinary feat, by a highly trained human athlete who is atypical for our species. By comparison, wild wolf packs can cover 30 miles of rough terrain in an hour (98), and migratory birds can cover 12,000 kilometers in 12 days, over open water without stopping for water or food (106). Comparative physiologists (74) and anthropologists (17) hold that traits of altitude tolerance and endurance running capacity are ancestral in humans, traceable to human origins in the high plains of east Africa. Today, however, the capacity for humans to survive at high altitude (136) and run down antelope in the bush may have more to do with strategy and the human capacity for temperature regulation than physical prowess.

In terms of the efficiency of walking and bicycling, investigators (36, 50, 51, 80, 107, 110, 132) commonly compute whole-body muscular efficiency to approximate 30%. That value for total body efficiency during steady-rate submaximal exercise conditions agrees closely with the results

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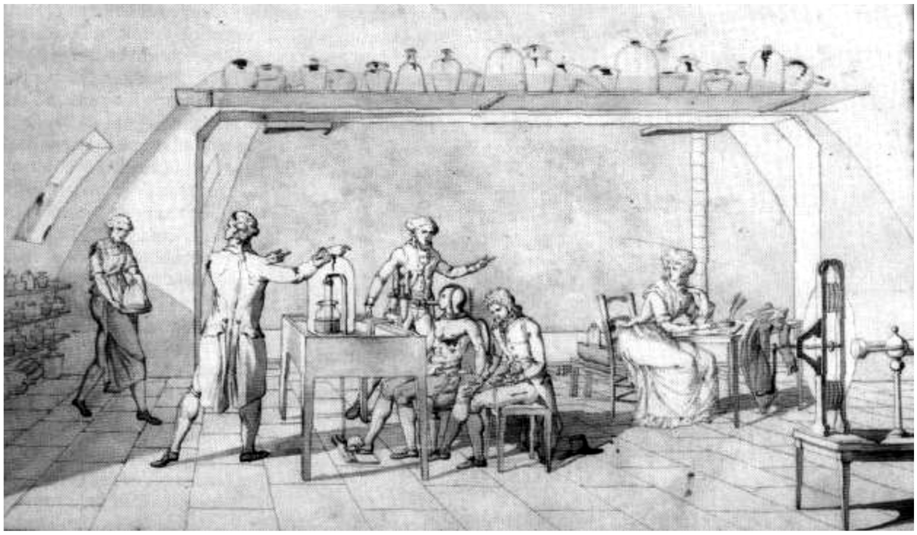

Figure 1 Antoine-Laurent de Lavoisier measures oxygen consumption on co-investigator Armand Seguin during foot treadle exercise, circa 1780. Drawings ascribed to his wife Marie Anne Paulze Lavoisier, who depicted herself at the table on the far right. The drawing is entitled "Expérience sur la respiration humaine" (Experiments into Respiration). Courtesy of the Division of Rare and Manuscript Collections, Cornell University Library.

from *ex vivo*-measured efficiencies of "phosphorylative-" (oxidative, 60%) and "mechanical-coupling" efficiencies (50%) [e.g., (0.60) (0.50) = 0.30]. The terms phosphorylative- and mechanical-coupling efficiencies are, respectively, used to denote the percentage of energy released in the catabolism of energy substrates captured as adenosine triphosphate (ATP), and the percentage of energy in ATP converted to mechanical work in the hydrolysis of ATP by interactions of the contractile proteins actin-myosin. The value of human locomotor efficiency during walking is superior to that of great apes (124), but is less than that of other bipeds such as kangaroos that are capable of harnessing kinetic energy from bound to bound (30).

In an elemental sense, the ability to undertake and sustain human muscular activity can be described in terms of the balance of energy demand and supply. Different forms of activity require (demand) different levels of energy which is supplied in the form of ATP. However, cellular levels of ATP are miniscule requiring real-time resupply of ATP used to do cellular work, always, and particularly in exercise when muscle power output and work of supporting systems rise more than an order of magnitude. With this understanding, power output exceeding the capacity of supply it cannot be sustained. As a first step in broaching this immense topic of human bioenergetics the power and capacity of energy systems in muscle are described.

Power and Capacity of Metabolic Energy Systems in Human Muscle

Knowing the sources of muscle energy has been an issue almost since the beginnings of modern biology. Indeed, Antoine-Laurent de Lavoisier, co-discoverer of oxygen, was the first to make determinations on resting and exercising humans (Fig. 1). Subsequently, it came to be known that as with unicellular organisms, isolated muscles could work in the absence of oxygen [for reviews see (22, 54)]. Such knowledge gave rise to the concepts of aerobic (oxidative) and anaerobic (nonoxidative) sources of energy. But, whereas in the early 20th century technology rapidly progressed in the ability to measure energy supply from oxidative metabolism in resting and exercising humans, it remained until the later part of that century and the advent of muscle biopsy and isotope tracer technologies to be able to estimate energy supply from nonoxidative energy systems. Still, to their credit knowing that muscles possessed oxidative and nonoxidative capacities for energy delivery, investigators in the early 20th century were resourceful in determining oxygen equivalents for the energy supplied by nonoxidative energy sources. One strategy was to compute the oxygen missed, that is, that which would have been needed to meet energy demand solely by oxidative metabolism. Typically computed for the rest to exercise transition, the oxygen missed was commonly termed the "oxygen deficit." Similarly, recognizing that exercise tasks of sufficient intensity to result in blood lactate accumulation were accompanied by a prolonged postexercise metabolic response, Hill and associates (67-70) developed the "oxygen debt" concept and used it in an attempt to compute an oxygen equivalent of the nonoxidative energy flux during exercise. Subsequently, with the discovery of the important role of phosphagen [ATP and phosphocreatine (phosphorylcreatine or PCr)], using the biexponential recovery O_2 debt curve Margaria and associates (95) segmented the recovery oxygen volume into two, "lactacid" and "alactacid" components. Although the terms are no longer in general use, it is important to know these terms, their derivations, intents and limitations (51).

Immediate energy sources in human muscle

ATP is utilized as the chemical energy source for actin-myosin interactions and most other endergonic, energy-requiring reactions in muscle such as Ca^{++} resequestration into the sarcoplasmic reticulum (SR) and plasma membrane Na^{+}-K^{+} exchange. There are three "immediate" or "phosphagen" energy sources available in skeletal muscle: ATP, PCr, and adenylate kinase (myokinase). Measuring the use of these energy storage forms is difficult during exercise for several reasons. The stores are finite, turnover is very rapid, and assessment by muscle biopsy and ^{31}P-MRS (magnetic resonance spectrometry) is challenging. Fortunately, however, in some respects Nature's design makes it possible to estimate energy flux rates and capacities. Even though ATP is the immediate energy source for actin-myosin interactions and related processes, the design of the muscle energy system is to maintain [ATP] homeostasis. Therefore, despite very rapid turnover, there is little net change in muscle [ATP] during physical exercise (16, 77, 119). Hence, whether muscle [ATP] is estimated by enzymatic analysis following muscle biopsy or NMR spectroscopy, the estimate of ATP storage capacity (5-6 mMol/kg wet weight) is small (<1.0 kcal/kg muscle) and the absence of net [ATP] change during exercise leads to discounting of ATP as a major muscle energy storage form.

In contrast to ATP content, [PCr] is five to six times greater in human muscle (16, 38), and [PCr] does decline during exercise in proportion to relative exercise intensity (REL) (16). Assuming a ΔG of –11 kcal/mol PCr, muscle [PCr] = 30 mMol/kg, and an active muscle mass of 30 kg in a 70 kg human, then the human muscle storage capacity of PCr approximates 10 kcal. Considering the capacity for human muscle to hydrolyze ATP during maximal efforts (45 kcal/min) (94), without replenishment, PCr would be exhausted within a fraction of a minute during hard muscular exercise (Table 1).

Table 1 Estimates of the power and capacity of human muscle energy systems

	Power (kcal/min)	Capacity (kcal)	Capacity oxygen equivalent (L)	Duration (s)
Immediate (phosphagen) (alactic)	45[a]	11[b]	1.4	15
Glycolytic (lactic)	22[a]	15	3	40
Oxidative	25[c]	1,200	240	∞[d]

Assumptions: 70 kg body weight, 30 kg active muscle.
[a]Power estimates from Margaria et al. (94).
[b]Phosphagen energy capacities from Edwards (38) and Karlsson et al. (84).
[c]VO_{2max} = 5 L/min.
[d]80% VO_{2max} sustainable for 60 min.

Adenylate kinase is expressed in human muscle and catalyzes the reaction 2 ADP → ATP + AMP. Adenosine monophosphate rises significantly in working human skeletal muscle (119), and through kinase activation AMP is thought to be an important signaling molecule stimulating both short- and long term metabolic adjustments. However, the energy yield from myokinase is small. Estimates vary (21, 38), but capacity is likely less than 1.0 kcal in 30 kg of human muscle.

Glycogenolysis and glycolysis

In human muscle, the phosphagen energy stores possess characteristics of high turnover, but low capacity requiring support from glycolytic and oxidative energy sources. During steady-state submaximal exercise, glycolytic and oxidative energy fluxes can be estimated from the combination of isotope tracer and indirect calorimetry techniques *(vide infra)*. Although significant lactate production occurs during steady-state exercise when muscle and blood lactate concentrations are constant, most is disposed of via oxidation, so the rate of oxygen consumption accurately includes and represents the energy contribution of "aerobic glycolysis." This means that during short-term, high intensity exercise the net rise in the body lactate pool represents the energy contribution of "nonoxidative," "anaerobic" glycolysis. Reasonable assumptions are that 1.0 ATP is produced per lactate anion accumulated when glucose is the precursor, and that 1.5 ATP is produced per lactate anion accumulated when glycogen is the precursor. As described below, from muscle biopsies and arterial-venous (a-v) difference measurements of net lactate release and lactate net accumulation can be estimated during nonsteady-state exercise (5). Alternatively, from the rise in blood lactate accumulation following maximal exercise efforts, and assuming that lactate mixes rapidly with total body water [an assumption now justified with the finding of ubiquitous expression of lactate (monocarboxylate) transporter (MCT) isoforms], nonoxidative energy production can be estimated. In the past measurements of "lactate tolerance" by Rodolfo Margaria, Paolo Cerretelli, Pietro diPrampero, Guido Ferretti and colleagues (the Milan School), have yielded estimated of 15 to 20 kcal energy capacity in a 70 kg human with 30 kg of human muscle (94). In agreement, others have produced similar estimates (21, 84). As with phosphagen energy stores, the capacities for glycogenolysis and glycolysis in human muscle are characterized by high turnover (> 20 kcal/min), but low capacity which means during maximal efforts the energy system is exhausted in 30 to 40 s such that continued efforts and recovery require support from oxidative energy sources.

Aerobic energy power and capacity

Compared to immediate and glycolytic energy production pathways, oxidative metabolism is denoted by slow response [$t/2 \approx 30$ s (29)], low to moderate power, but very large capacity. In overnight fasted humans habituated to a mixed diet, either resting or engaged in prolonged, submaximal exercise eliciting a pulmonary respiratory exchange ratio (RER = VCO_2/VO_2) of 0.82, approximately 4.82 kcal are derived per liter O_2 consumed. From a resting baseline of approximately 2.6 to 3.3 ml/kg body mass per minute (27), resting metabolic rate approximates 1.0 kcal/min with no appreciable contributions from phosphagen pools or anaerobic glycolysis. From a resting baseline, peak aerobic energy production will depend on maximum oxygen consumption (VO_{2max}), such that a person capable of raising their resting VO_2 20-fold will have twice the capacity for aerobic metabolism (i.e., 20 multiples of resting metabolism, METS) compared to someone capable of only 10 METS. Further, a person with a large MET capacity will typically accomplish a given exercise task with a lower pulmonary RER, and presumably lower muscle respiratory quotient (RQ), than a person with a lesser aerobic capacity while exercising at the same power output. Consequently, an individual with a greater aerobic capacity will derive a lower percentage of energy from CHO-derived fuel sources (glycogen, glucose, and lactate), and a greater percentage of energy from lipid energy sources [plasma FFA and intramuscular triglycerides (IMTGs)] compared to an individual with a lesser aerobic capacity.

Energetics of isolated muscles and muscle fibers

As described by Mommaerts (100), early muscle physiologists such as Hill (65) used thermopiles to measure the energies (heats) of muscle activation and shortening in nonperfused frog muscles stimulated to contract at room temperature. Over decades the device was improved (66), and subsequently initial (contraction) and latent (recovery) heats could be observed not only in amphibian muscles, but also in mammalian muscle bundles. Ultimately, initial heats could be described as representing separate heats of activation and shortening (100). Activation heat is measured during isometric or isotonic contractions and is now accepted to be associated with ATP hydrolysis attributable to ion pumping, specifically the cell membrane Na^+-K^+-ATPase and, quantitatively more importantly, the Ca^{++}-ATPase associated with calcium ion resequestration into the SR (43, 100).

In his initial experiments, Hill (65) observed that if muscles were allowed to shorten additional heat was released; this was termed "shortening heat" and subsequently became associated with the work done (42), and now ascribed to myosin-actin cross-bridge interactions in addition to those used for isometric contraction (100). Fast muscle fibers are distinguished by unique myosin isoforms that have high ATPase activity and rapid cross-bridge turnover (6-8). As well, fast fibers are arranged in larger motor units, innervated by larger, more heavily myelinated alpha motor neurons with more extensive motor end plates (8). Given its design features, there is every reason why fast fibers contract and relax more rapidly than do slow fibers. As well, there is considerable evidence that the price of speed in fast muscle is a loss of economy (123).

Human Muscle Power Output and Oxygen Consumption *In Vivo*

In the process of oxidative phosphorylation, working muscle consumes oxygen and produces carbon dioxide. However, measuring the metabolic response of human muscles to gradations in power output is not trivial. In the past, most measurements were whole-body (pulmonary) oxygen consumption, *vide infra* (Figs. 16-18). In using pulmonary gas exchange to reach conclusions about working muscle investigators assume that working muscle dominates the whole-body metabolic response. In retrospect, that assumption proves to be correct because of the extraordinary and innovative efforts of a few investigators who have employed combinations of pulmonary and muscle measurements (12, 103, 134). More recent studies of muscle metabolic measurements have involved determinations of indirect calorimetry and NMR spectroscopy (89).

Assessments of muscle respiration by simultaneous pulmonary and working muscle measurements

Working leg muscle oxygen-consumption rates are rare due to the invasiveness and difficulties in measuring blood flow and femoral arterial and venous oxygen and carbon dioxide contents. Still more rare in the literature are measurements in which leg respiratory quotient (RQ = VCO_2/VO_2) were determined over a range of power outputs so that working muscle efficiency could be determined. An example of a study in which pulmonary and working leg muscle rates of oxygen consumption were determined simultaneously over a range of exercise intensities (leg ergometer cycling) is that of David Poole, Peter Wagner and colleagues (107) (Fig. 2). Results of that study were subsequently reproduced (47). In Poole et al., from the inverse of the regression of the caloric equivalent of pulmonary VO_2 on the caloric equivalent of external power output during leg cycling muscular efficiency determined from pulmonary gas exchange averaged 29.1 $\pm$ 0.6% (95% CI 27.9-30.3%). Simultaneously, muscle efficiency calculated from the leg VO_2 averaged 33.7 $\pm$ 2.4% (95% CI 29.0-38.4%) (Fig. 2). If for these experiments the phosphorylative coupling efficiency is taken to be 60%, then the mechanical coupling efficiency for human leg muscle is 56%.

In the above example, the confidence intervals for muscle are larger than those for pulmonary VO_2. Among the technical problems associated with the muscle are catheter placement (ante- or retrograde), the admixture of blood from nonmuscular tissues or auxiliary muscles, and determination of arterial and venous oxygen contents. In the example cited, blood CO_2 contents were not measured, and so a RQ of 0.95 as seen in previous studies and subsequently reconfirmed by us (12),

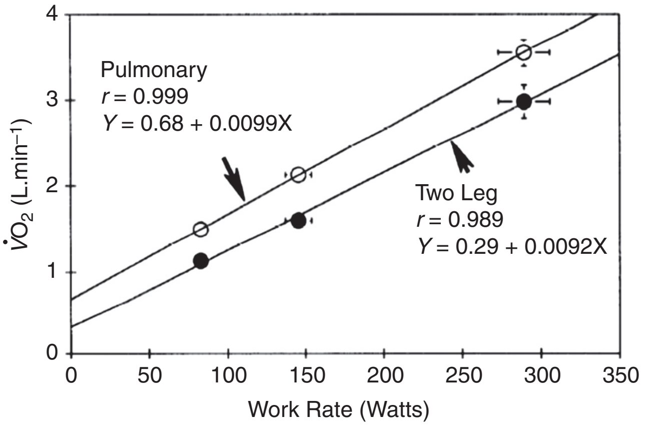

Figure 2 Group mean values (± SE) of studies on 17 men showing linear and parallel increments in pulmonary and working muscle (leg) oxygen uptake. Pulmonary and leg VO_2 measurements yield delta efficiencies of measurements (29.1 ±0.6%) and (33.7 ± 2.4%) for whole body and working muscles, respectively. The difference between y intercepts shows that the body provides significant metabolic support for processes outside the exercising legs, but that these metabolic "costs," such as the work of breathing and gluconeogenesis, change in proportion to muscle power output. From Poole et al. (107) and used with permission.

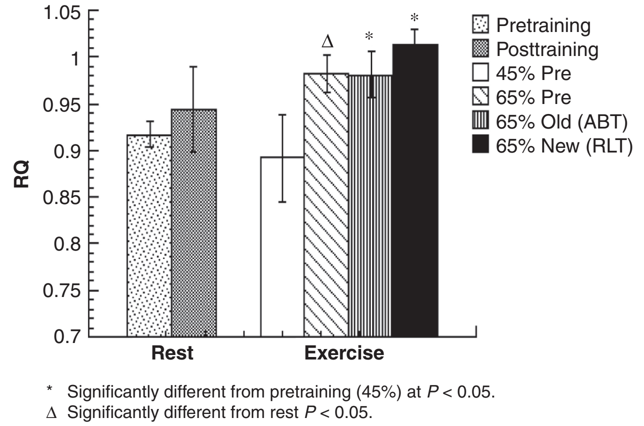

Figure 3 Working muscle respiratory quotient (RQ) as determined by femoral arterial and venous CO_2 and O_2 concentration difference measurement. Values are means ± SEM for eight subjects. Exercise data are means of last 30 min of exercise. Subjects studied 2× before training (i.e., @ 45% and 65% VO_{2peak}), and twice after training (i.e., ABT = same absolute intensity as 65% pretraining, and RLT = same relative intensity as pretraining, i.e., 65% VO_{2peak}). *Significantly different from pretraining (45%) ($P < 0.05$); $^{\Delta}$significantly different from rest, $P < 0.05$). Results show working muscle to be carbohydrate dependent, both before and after training. From Bergman et al. (12) and used with permission.

was assumed to compute muscle RQ and the caloric equivalent of VO_2. Considering literature values for working muscle RQ (12, 47, 103), the assumption of an RQ of 0.95 was appropriate, especially for higher power output exercises. As a result of parallel slopes in pulmonary and working muscle respiratory in responses to muscle power output requirements it can be concluded that human muscle contracts with an overall efficiency on the order of 30%. Again, that value for an overall efficiency agrees well with *ex vivo* estimates of 50% for mechanical coupling efficiency and 60% for phosphorylative coupling efficiency, respectively. The efficiency and economy of human muscle exercise are discussed below.

A pulmonary respiratory exchange ratio—working muscle respiratory quotient paradox?

By classic as well as contemporary standards measuring pulmonary carbon dioxide production, hence RER (or R = VCO_2/VO_2) is a standard technique well within the technical capabilities of most clinical and research laboratories. However, measuring arterial and venous O_2 and CO_2 contents as well as tissue (usually working limb) blood flow is technically demanding. Of the parameters to be measured, the CO_2 content of arterial and venous blood is perhaps most complex, and not always attempted, as in Poole et al. (107) (Fig. 2). Still, knowing the "combustion coefficient" of working muscle provides invaluable information on the energy-substrate partitioning in working muscle. From the first law of thermodynamics (principle of conservation of mass) one would anticipate that over time RER and RQ would be equivalent. However, in two investigations, we (12, 47) (Fig. 3) found working muscle RQ to be slightly higher than pulmonary RQ. The differences are small (0.03), but consistent suggesting relatively greater CHO, and less lipid oxidation in working muscle as compared to the whole body. In our studies, glucose (13, 47, 48) and lipid (12, 47, 48, 79, 129) whole-body and working muscle fluxes were measured by the combination of tracers and limb net balance techniques. Like the results of others (112) our measurements showed that most (≥90%) of glucose disposal (Rd) was accounted for by net uptake by working muscle (13). As well, we could not observe a net change in muscle IMTG content by muscle biopsy analysis (12, 49, 129), and net releases of glycerol (a surrogate for intramuscular lipolysis) was small. Overall, these results support the notion of a shunt of available carbohydrate energy stores to working muscle with a relatively greater reliance on lipid energy sources by the remainder of the body during muscle exercise (12, 13).

Classic Compared to Theoretical-Thermodynamic Approaches to Estimating Energy Yield from Substrate Oxidation

Working muscle oxygen consumption, RQ, P/O, and energy yield

There are two approaches to solving the problem of determining the energy yield per unit of oxygen consumed. The classic approach is to utilize indirect calorimetry to determine pulmonary VO_2 and VCO_2, and then to use standardized tables to determine the energy expenditure in kcal/min (or kJ). A more recent approach is to use indirect calorimetry to determine VO_2, and from there calculate the ADP to ATP phosphorylation rate using assumptions from contemporary biochemistry; the latter is referred to as the "theoretical-thermodynamic approach," a term first used by Whipp and Wasserman (132) and adopted by others (50, 80). The two methods, classic and theoretical-thermodynamic, yield very similar results, with advantages being in favor of the classic method in terms of ease of use, particularly when employing the RER for evaluating effects of variables (nutrition, exercise, training, gender, aging, and environment) on energy substrate partitioning.

The classic approach to determining metabolism

In classic metabolic biochemistry based on the cumulative efforts of Antoine-Laurent de Lavoisier, John Scott Haldane (57), Francis Benedict (9, 10), C. Gordon Douglas (37), A. Monmouth Smith (118), Wilbur Olin Atwater (1), Edward Bennett Rosa (1, 2), and Nathan Zuntz (137) and their colleagues, if the pulmonary RER equals 1.0 (RER $= VCO_2/VO_2 = 1.0$) carbohydrate is the fuel (e.g., for glucose: $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$, RQ $= VCO_2/VO_2 = 6/6 = 1.0$) and the energy yield from bomb calorimetery (81) is 5.05 kcal/L O_2. If on the other hand, the pulmonary RER approximates 0.71 (e.g., for the most common triglyceride Trioleate: $C_{57}H_{104}O_6 + 80\ O_2 \rightarrow 57\ CO_2 + 52\ H_20$, RQ $= 57/80 = 0.71$), lipid is the fuel and the energy yield is 4.69 kcal/L O_2. As originally described by Zuntz and Schumburg (137), and subsequently modified by Lusk (93), the relative contributions of lipids and carbohydrates to the fuel energy source and their contributions to energy flux could be interpolated for RER values between 1.0 and 0.71. Consequently, using CHO combustion as the standard, 7.7% more energy is liberated per unit oxygen consumed if CHO is the fuel compared to lipid. Hence, to a metabolic biochemist the advantage of CHO combustion is obvious when oxygen supply is limited.

To a metabolic biochemist (73) or physiologist (20,21,49), the advantage of lipid oxidation is revealed when the enthalpy/unit weight of lipid is considered with a yield of 9.1 kcal/g (38 kJ/g, 1 kcal = 4.19 kJ) as opposed to a value of 4.2 kcal/g (18 kJ/g) for CHO oxidation. Hence, given the 2.17 ratio of energy per gram of lipid compared to energy per gram of carbohydrate, lipid yields 117% greater yield per unit weight making lipid the preferred energy storage form in biology. The superior energy yield of lipids over carbohydrate is further amplified because glycogen is hydrated *in vivo*. Estimates vary, but a value ≥ 2.7 g H_2O/g glycogen is widely used (116). Superiority of the energy density (compactness) of lipid over glycogen energy storage is further emphasized when the energy/mol equivalent is computed. For an average triglyceride, 860 g/mol is typically assumed (12, 454, 46, 129); so the energy yield is (9.1 kcal/g $\times$ 860 g/mol =) 7,826 kcal/mol. In contrast, for glucose (180 g/mol), the energy yield is (4.2 kcal/g $\times$ 180 g/mol =) 756 kcal/mol, or one-tenth that of lipid.

Theoretical-thermodynamic approach to determining metabolism

In contrast to the use of standard tables derived at the end of the 19th century, and modified early in the 20th century used by metabolic biochemists and physiologists, a structural biochemist's accounting involves determining the ATP yield per unit of substrate oxidized. For glucose oxidation ($C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$), and depending on the cytoplasmic to mitochondrial redox shuttle used (i.e., glycerol-phosphate, malate-aspartate, or lactate shuttle), the ATP yield from glucose oxidation will be 36 to 38 mol ATP/mol glucose (91), making the P/O (inorganic phosphate, or ADP phosphorylated, per atom of oxygen consumed) for glucose oxidation $\approx 36/12$ to 38/12, or 3.0 to 3.17. This is because the malate-aspartate and lactate shuttles are NADH-linked, and the P/O for NADH is 3.0 (90), whereas the glycerol-phosphate shuttle uses FADH2 as the reducing equivalent (hydride ion) carrier, with the P/O for FADH2 is 2.0. If NADH-linked cytoplasmic to mitochondrial lactate shuttles are assumed (P/O=3), then for glucose an ATP yield of 38 ATP/mol glucose and ΔG of -11 kcal/mol ATP yields -418 kcal/mol glucose, or 418 kcal/180 g, or 2.32 kcal/g effectively trapped in the form of ATP from aerobic glycolysis with a P/O of 3 to 3.2.

Similarly, for the oxidation of a typical fatty acid such as palmitate (16 C), a structural biochemist would calculate that the ATP yield from palmitate oxidation to be 129 mol ATP per mol palmitate. In palmitate oxidation ($C_{16}H_{31}O_2 + 23\ O_2 \rightarrow 16\ CO_2 + 145\ H_2O$, RQ $16/23 = 0.70$), with one ATP (yielding AMP) used for activation, seven cycles of the β-oxidation pathway will yield 7 NADH and 7 FADH2, or (3+2=) 5 ATP/β-oxidation cycle, or 35 ATP from β-oxidation cycling associated with the catabolism of a single palmitate molecule. As well, each β-oxidation cycle yields an Acetyl-CoA; each TCA cycle yields 3 NADH, 1 FADH2, and 1 GTP ($\approx$ 1 ATP), or 12 ATP/cycle; so TCA cycle activity from palmitate yields 96 ATP associated with the catabolism of a single palmitate molecule. Hence, for palmitate the yield is 96 ATP from β-oxidation + 35 ATP from tricarboxylic acid cycle (TCA) cycle activity – 2 ATP for activation =) 129 ATP with an overall P/O (129 ATP/46 O) = 2.8. In terms of energy from the oxidation of palmitate, and ATP yield of 129 mol/mol palmitate and a ΔG of -11 kcal/mol ATP, the energy yield from palmitate is $-1{,}419$ kcal/mol palmitate, or 1,419 kcal/256 g, or 5.54 kcal/g effectively trapped in the form of ATP from the oxidation of 1 g of palmitate, again with a P/O of 2.8. As implied from the greater energy yield of CHO compared to lipid oxidation (5.05 vs. 4.69 kcal/L O_2, a 7.8% difference), the P/O from the oxidation of carbohydrate-derived oxidative catabolism (3.0 vs 2.8 mol ATP/mol atom oxygen) represents a 7.1 % difference.

Before leaving this section, it is appropriate to note the conclusion of a greater energy yield per atom of oxygen for CHO dervied, as opposed to lipid-derived substrates from classic indirect calorimery and the stoichiometry of biochemistry is also supported by studies on isolated mitochondria (18). However, because of shattering of the mitochondrial reticulum during isolation of mitochondrial vessicles, proton leaks across inner mitochondrial membranse are significant to the extent the P/Os are affected, making the difference between energy yields from lipid- and CHO-derived fuels 15% per atom of oxygen. Nonetheless, it is clear that regardless of method, CHO oxidation provides more energy when oxygen supply is limited, such as during hard exercise. Energy yield per gram of lipid is, however, far greater than for CHO oxidation in a unit mass basis (*vide supra*).

Whether by classic or theoretical-thermodynamic approaches there is no evidence that exercise training improves the mitochondrial or glycolytic P/O, which represents the efficiency by which the chemical potential energy in dietary-derived energy substrates is converted to ATP. However, training does raise the maximum rate of oxygen consumption and decreases the RER at which a given submaximal power output can be accomplished. As well, the learning of complex neuromuscular tasks may improve the economy of movement, and in the aggregate these contribute toward the improvements in performance associated with training. However, as is currently known, the P/Os for CHO- and lipid-derived substrates approximate 3 and 2.8, respectfully. If training shifts energy-substrate partitioning to greater lipid use, then the measured VO_2 to accomplish a task may rise slightly.

Protein and amino acids as fuels

Dietary proteins provide essential building blocks to body structures, whereas dietary carbohydrates and lipids serve to provide the energy for diverse body functions including protein synthesis. All three classes of dietary energy sources require digestion, assimilation, distribution, and cellular uptake prior to entry into final common metabolic pathways of catabolism. In the process of becoming energy sources, amino acids require one additional step, nitrogen removal, that is accomplished by trans- or deamination. Hence, in terms of the time from assimilation, and the metabolic power of the three classes of dietary energy sources the hierarchy of substrate use is CHO > lipid > amino acids.

Early on in the study of human nutrition, it became obvious that physical exercise had little effect on urinary nitrogen excretion (28, 86, 87). Subsequently, in detailed studies of total body nitrogen balance involving strict dietary controls and the collection of urinary, fecal, and sweat nitrogenous products Todd et al. (126) established that so long as dietary energy was adequate to cover need, physical activity did not result in negative nitrogen balance. Subsequently, with knowledge that little additional urinary nitrogen is excreted because of exercise, and because of the inconveniences and difficulties in measuring changes in nitrogen balance associated with physical exercise, for studies of energy substrate partitioning during exercise, investigators (e.g., 11, 128) have not bothered to correct pulmonary RER measurements for amino acid and protein use during exercise. By default then, the general view became that carbohydrate- and lipid-derived fuels are preferred energy substrates for exercise.

With the advent of isotope tracer methodology several sets of investigators evaluated the effects of various stresses on amino acid metabolism in muscle preparations (102), intact rats (61, 133), and intact functioning humans (99, 135). Such studies forced recognition that the oxidation of some classes of amino acids, such as branched-chain amino acids, increase during exercise. However, it is also apparent that while the oxidation of ketogenic amino acids such as leucine scales to metabolic rate during exercise, overall total flux changes little, with increased disposal via oxidation compensated for by decreased non-oxidative disposal (99).

So far as the effect of exercise training on oxidative disposal of key amino acids is concerned, from earlier studies on trained and untrained rats (61) it appeared that training increased oxidative disposal of leucine during exercise. The result made sense as training increased muscle mitochondrial mass (35, 75). However, in studies on men and women studied both before and after endurance training McKenzie et al. (97) showed that training suppressed leucine oxidation whether measured at given absolute or relative intensities, especially in young women who were also capable of greater lipid oxidation than their male cohorts.

While exercise has minimal effects on amino acid flux rates and nitrogen balance, it remains that amino acids and proteins play important roles in sustaining physical exercise; but, of those roles, that of an energy substrate during exercise is perhaps the least important. Still, the assumption that zero amino acid oxidation occurs during exercise is unjustified and a matter of concern. Estimates of the roles played by amino acids as fuel sources during exercise vary from 4% to 10%, but there is little security in those estimates as individual amino acids behave differently and tracing one gives little confidence about what is happening to the others assessed by leg RQ or pulmonary RER determinations. This situation is better in estimating plasma FFA oxidation, as one fatty acid (e.g., palmitate or oleate) can be traced, and knowing its absolute and relative concentrations as well as its flux and oxidation rates, the assumption can be made that what happens to the traced plasma FFA is representative of the total.

Another approach to estimating the relative role of proteins and amino acids to the fuel energy source can be made from dietary records (79). From urinary nitrogen excretion measurements or food records obtained on well-nourished and weight stable individuals the contribution of amino acids and proteins to daily energy expenditure can be made. For instance, if dietary proteins and amino acids supply 15% of daily energy intake, then the contribution of proteins and amino acids to pre-exercise energy expenditure can be assumed to be 15%. Next, assuming that total amino acid oxidation

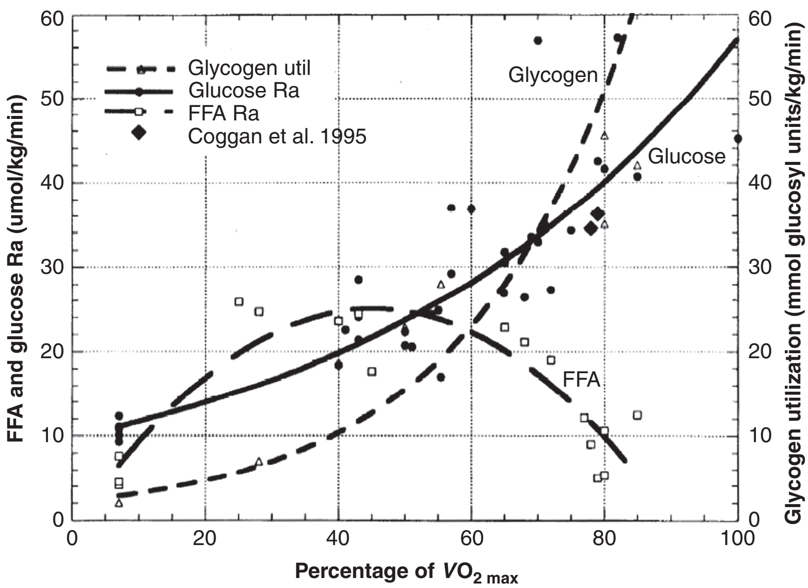

Figure 4 Results of an extensive literature search showing blood glucose and free fatty acid flux rates (Ra) and net muscle glycogenolysis (77) as functions of relative exercise intensity (REL) as given by % VO_{2max}. This form of analysis indicates exponential increments in muscle glycogenolysis and glucose Ra as functions of relative exercise intensity. In contrast, the analysis shows multicomponent polynomial response of plasma FFA flux, with easy to moderate intensity exercise (i.e., 25%-40% VO_{2max}) eliciting a large rise in flux, but crossover and decreasing flux at approximately 50% VO_{2max}. Note that plasma free fatty acid (FFA) flux is predicted to reach minimal values as $VO2_{max}$ is approached. For glycogen utilization, $y = 2.11\ e(0.04\times)$, $r^2 = 0.87$; for glucose Ra, $y = 9.8\ e(0.02\times)$, $r^2 = 0.84$, and for FFA Ra, $y = -0.833 + 1.14\times - 0.013\times 2$, $r^2 = 0.87$. From Brooks and Trimmer (24) and used with permission.

remains constant during exercise in which total energy expenditure rises, perhaps 15-fold, then amino acids contribute 1% of total energy expenditure. If, on the other hand, amino acid oxidation rises 50 % (97) or 400% during exercise (99), estimates are that amino acids contribute 2% to 4% of total energy substrate, which has the effect of lessening the calculated lipid oxidation by a corresponding amount. Hence, as originally suspected from the lack of effect on urinary nitrogen excretion, the fuel energy role of amino acids is relatively small in well-nourished individuals, with the result being that the total body RER is a reasonable estimate of the nonprotein RQ during exercise, and a correction for amino acid oxidation is typically not used even when available (79).

Exercise and Energy Substrate Partitioning in Humans Crossover Effects

Over the last several decades both cross-sectional and longitudinal training studies to determine effects of gender, exercise and exercise training on lipid metabolism in young men and women have been conducted. For example, in our studies we have studied subjects engaged in moderate (45%-50% VO_{2peak}) and hard (65% VO_{2peak}) intensity exercises during which rates of O_2 consumption and CO_2 production are submaximal and constant (steady), both before and after 10 to 12 weeks of endurance training. This design permits pre- and posttraining comparisons to be made at given absolute power outputs and relative exercise intensities. On the basis of those investigations as well as data from a variety of sources, it is clear that results follow predictions of the Crossover Concept (23) (Fig. 4) such that CHO oxidation predominates in working muscle and at the whole-body level and that lipid is used sparingly in working skeletal muscle, especially if exercise intensity is greater than 65% VO_{2max}, or if exercise duration is short (12, 49, 63, 64, 82, 103, 129).

While developed on humans, (Fig. 4) the Crossover Concept appears applicable to other mammalian species with data obtained on dogs goats and rats all seeming to follow the human pattern (Fig. 5) (113). In the figure shown, given a composite of lipid and CHO oxidation rates across ranges of relative exercise intensities, the comparisons between dogs and goats are interesting. The animals were of similar body masses, but whereas running goats have aerobic capacities like those of humans (40-60 ml/kg/min VO_{2max}), running dogs have much higher aerobic capacities, similar to those of rats (80-100 ml/kg/min VO_{2max}). Still, despite significant differences in aerobic capacities and body sizes, when expressed relative to VO_{2max} the balance of lipid and CHO use is well preserved across mammalian species.

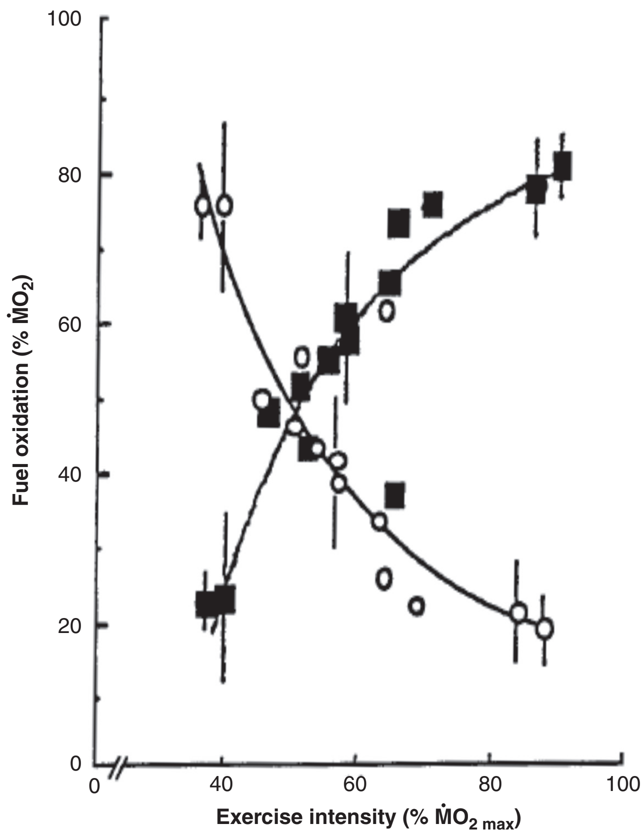

Figure 5 Effect of exercise intensity on the balance of lipid (○) and carbohydrate (■) oxidation in four mammalian species (goats, dogs, rats, and humans). Mean (± SEM) data on dogs and goats are shown in the center and quadrants; other data on rats and humans have been included from the literature. Regardless of body size and configuration or aerobic capacity, the same patterns of energy substrate partitioning during physical activity are apparent. In this respect the data are to be compared to those in Figure 1. Redrawn, with permission, from two separate figures in Roberts et al. (113) by GAB.

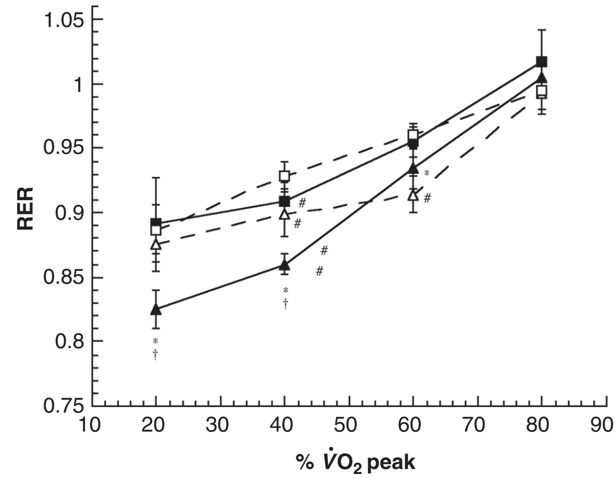

Figure 6 Relationship between respiratory gas exchange ratio RER = VCO_2/VO_2 in trained (T) and untrained (UT) men during sustained exercise. Subjects were studied in fed and over night fasted (postabsorptive) conditions. Trained and fasted individuals have the lowest RERs, but only during easy to mild-intensity exercise. Overall, results show predominance of carbohydrate (CHO) over Lox during exercise regardless of training state or dietary condition. From Bergman and Brooks (11) and used with permission.

Exercise, exercise training, and preexercise nutrition effects on whole body energy substrate partitioning as assessed from RER determinations

By two actions endurance training increases the ability to utilize lipid energy sources. First, training raises VO_{2max}, thereby allowing men and women to perform a given task at a lower relative intensity (as given by % VO_{2max}). Second, by increasing mitochondrial mass, endurance training increases the sensitivity of respiratory control by raising cytosolic adenylate energy charge (ATP/ADP) and redox status ($NADH/NAD^+$), thus downregulating glycolysis and relieving its inhibitory effects of mitochondrial FFA uptake and oxidation (19, 119). At the whole body level, measurements of pulmonary O_2 and CO_2 exchange allow computation of the whole body RER ($=VCO_2/VO_2$), which in turn allows assessment of the relative contributions of CHO and lipid to the total fuel mix under steady-state conditions. To assess effects of exercise intensity, training, and recent nutrition on energy substrate partitioning, we (11) (Fig. 6) studied seven untrained men and seven category two cyclists during graded exercise from 20% to 80% VO_{2peak} after a day of rest to normalize muscle glycogen levels. To assure steady pulmonary gas exchange and blood lactate values, measurements were made over periods of 45 min (80% VO_{2peak}), 90 min (60% VO_{2peak}), and 120 min (20% and 40% VO_{2peak}). Subjects were studied after either a 12-hour overnight fast, or 3 h after a 500 kcal (53% CHO, 2% fat, and 16% protein), low glycemic index breakfast. Effects of exercise, prior endurance training, and recent nutrition were evident. Training decreased RER (increased lipid oxidation) most notably at low relative power outputs, especially when trained subjects were studied overnight fasted. However, when relative power output increased from mild (20%-40% VO_{2peak}) to moderate (60% VO_{2peak}) and hard (80% VO_{2peak}) exercises, RER increased to values close to unity indicating predominance of CHO oxidation in all subjects regardless of training state or dietary condition (Fig. 6).

Exercise and exercise training effects on muscle energy substrate partitioning as assessed leg respiratory quotient determinations

Endurance training raises fat metabolism during exercise by allowing men and women to perform a given task at a lower % VO_{2max}. For example, in a longitudinal training study on nine men, by means of femoral arterial and venous catheterization and limb blood flow measurements we determined glycerol, FFA, glucose, and lactate exchanges across the working limb. Femoral and arterial blood samples were taken for determinations of CO_2 and O_2 contents so that limb RQ, and the intramuscular balance of CHO and lipid oxidation could be determined. As well, muscle biopsies were taken before and after 1-h exercise bouts to determine net changes in muscle glycogen and IMTG contents could be compared with CHO and lipid oxidation rates determined from pulmonary gas exchange and leg RQ determinations. Men were studied during 1 h of cycle ergometry at two intensities before training (45% and 65% VO_{2peak}) and after training [65% pretraining VO_{2peak}, same absolute workload (ABT), and 65% posttraining VO_{2peak}, same relative intensity (RLT)]. Supervised training involved nine weeks of leg cycle ergometry (1 h/day, 6 days/week @75% VO_{2peak}). Over the course of a given exercise bout, arterial pH and [lactate] values were constant (12, 49) suggesting that a-v blood gas values would reflect tissue substrate utilization. Prior to training, during exercise at 45% VO_{2peak} leg RQ was 0.89±0.05 (Fig. 2). Otherwise, during the 65% VO_{2peak} trials leg RQ was in the range of 0.95 to 1.0. These data (Fig. 3) indicate little lipid oxidation in the working leg. Very similar results have been obtained by Odland et al. in Heigenhauser's group (103). As with results of the cross-sectional study (Fig. 6), results of our longitudinal study indicate that working muscle oxidizes predominantly CHO, even after months of endurance training. Data in Figure 3 are consistent with other results obtained in the same study such as the absence of glycerol release (Fig. 11), very small net FFA uptake, and no net decrease in IMTG content (12, 129). In the aggregate, these results indicate limited lipid oxidation or IMTG mobilization in working skeletal regardless of training state.

Effects of exercise and exercise training on blood glucose and glycogen use

Plasma glucose use (rate of disposal or disappearance, Rd) is significant in resting postprandial individuals, but glycemia is maintained because hepatic glucose production (rate of appearance, Ra) is equivalent. Depending on time since last eating, glucose is a major source of the carbohydrate oxidized in resting individuals. As seen Figure 7A (48), endurance training has no significant effect of glucose Rd (or Ra) in resting individuals. However, both before and after training, glucose disposal rises as an exponential function of REL as given by $\%VO_{2max}$ (Fig. 7B) (48).

Not only does glucose flux rise during exercise, but also endurance training has significant effects on glucose use, as measured in Rd and reflected also in Ra. Seen in the comparison between second and third exercise histogram bars (Fig. 7A), 10 to 12 weeks of endurance training decreases glucose flux for a given absolute exercise power output (ABT), in this case that which elicited 65% of VO_{2max} before training (65UT) and 52% of VO_{2max} after training. However, as seen by comparing the fourth with other exercise histogram bars, training increases glucose flux for a given REL. After training, the REL task at 65% of posttraining VO_{2max} elicited a 43% greater than that during exercise at 65% of VO_{2max} pretraining (65UT) reflecting a significantly greater capacity for hepatic glucose production to maintain glycemia.

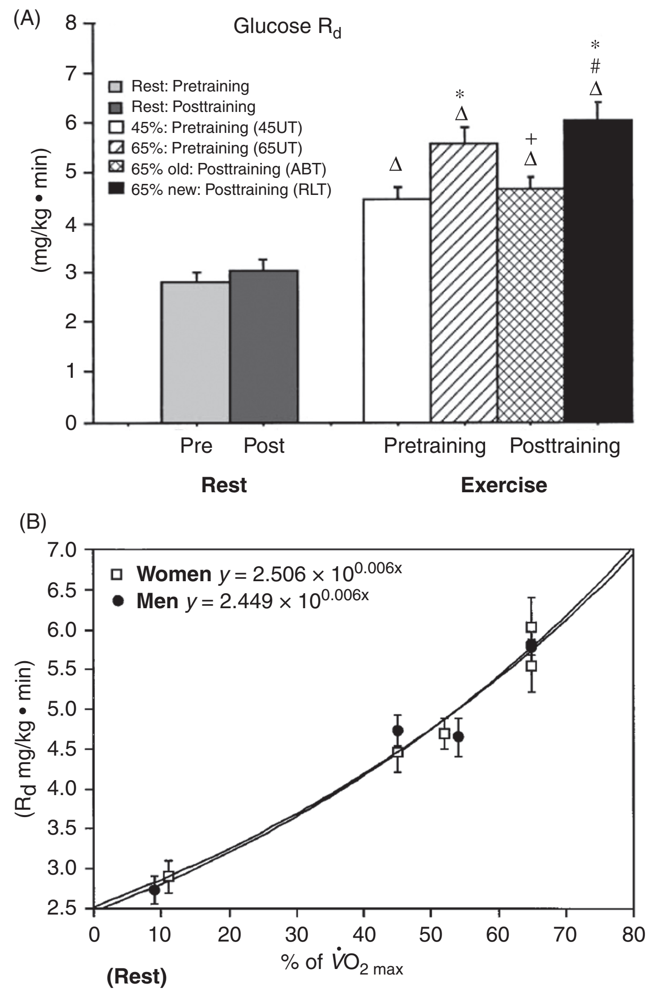

Figure 7 (A) Effect of exercise intensity and training on the plasma glucose rate of disappearance (Rd). Values are means ±SE of last 15 and 30 min for rest and exercise, respectively, for 17 women. (B) Relationship between glucose rate of disappearance (Rd) exercise intensity as given by $\%VO_{2max}$ in 19 men and 17 women, before and after 10 to 12 weeks of endurance training. Note the exponential rise in glucose use as a function of exercise power output. Values are means ± SE; $^{\Delta}$significantly different from rest, $P < 0.05$. *Significantly different from 45UT (untrained), $P < 0.05$. From Friedlander et al. (48) and used with permission.

In normal sized adults, with a blood volume approximating 5 L and a blood [glucose] of 100 mg/dL, the circulating blood glucose pool amounts to a diminutive 5 grams, or roughly 21 kcal. Consequently, while exercise significantly raises blood glucose use (Fig. 7A and B), whether at rest or during exercise, glucose provides only a small portion of the carbohydrate energy used, and even a smaller part of the total energy used when contributions of muscle glycogen and lipid energy sources are considered (Fig. 8) (48). Hence, while glucose use scales exponentially to REL (Figs. 4 and 7B), the gain is small compared to general rise in metabolic rate. This mechanism of a restraint on blood glucose use during exercise points to the need to protect glycemia for cerebral metabolism under all conditions.

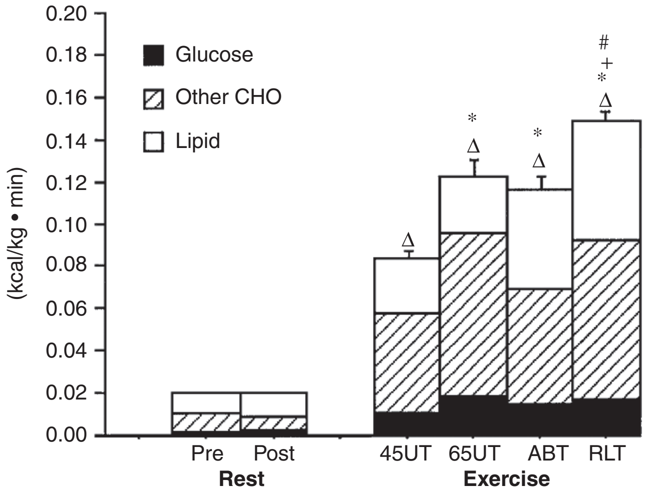

Figure 8 Contributions of energy from different substrate sources during rest and exercise before and after training. SE and statistical symbols are for total energy expenditure only. Values are for nine subjects. CHO, carbohydrate. $^{\Delta}$Significantly different from rest; *significantly different from 45UT; $^{+}$significantly different from 65UT; and $^{\#}$significantly different from ABT, $P < 0.05$. From Friedlander et al. (48) and used with permission.

Effects of exercise and exercise training on the use of plasma free fatty acids and other lipid energy sources

On arising after 8 h of sleep and having eaten 10 to 12 h previously, lipids comprise most of the energy utilized (Fig. 6). Of the lipid energy sources used, free fatty acids derived from adipose triglycerides, as well as other lipid energy sources are used. As seen in Figure 9, in individuals adequately nourished, endurance training has no significant effect on resting FFA Rd (or Ra). A similar statement can be made about glycerol flux (a marker of adipose lipolysis) in resting exercise trained and untrained individuals. However, both before and after training, FFA use (Rd) rises during easy to moderate REL as given by %VO_{2max} (Fig. 9). However, as shown in Figures 4 and 9, FFA Rd does not rise exponentially as does glucose flux, but rather FFA flux and oxidation as well as total lipid oxidation rates are described as inverted hyperbolas. As shown in Figure 9A and B for women (46) and men (45), respectively, prior to 12 weeks of endurance training FFA Rd increases significantly during easy, 45% VO_{2max} exercise (45UT). However, as seen by comparing the first and second exercise histogram bars, in untrained individuals FFA Rd is less during hard 65% VO_{2max} exercise (65UT) compared to that during easy to moderate intensity exercise at 45% VO_{2max} (45UT).

Not only does FFA flux rise during exercise, but also endurance training has significant effects of FFA mobilization and use in young women, as measured in Rd and reflected also in Ra. Seen in the comparison between second and third exercise histogram bars (Fig. 9A), in young women 10 to 12 weeks of endurance training increases FFA flux for a given absolute exercise power output (ABT). In women, as seen

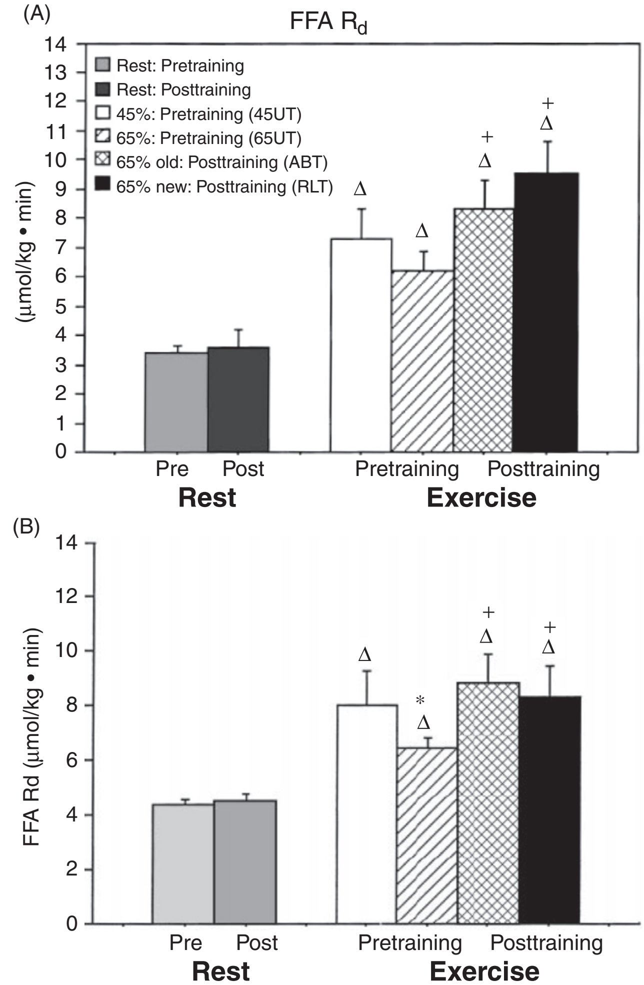

Figure 9 (A) Effect of exercise intensity and training on plasma FFA Rd. Values are means 6 SE of the last 15 and 30 min for rest and exercise, respectively; $n = 8$ young women. $^{\Delta}$Significantly different from rest; *significantly different from 45UT; $^{+}$significantly different from 65UT; and #significantly different from ABT ($P < 0.05$). From Friedlander et al. (46). (B) Effect of exercise intensity and training on plasma FFA rate of disappearance (Rd) in 10 young men before and after 10 weeks of supervised endurance training. Values are means ± SE of the last 15 and 30 min for rest and exercise, respectively; $n = 9$ subjects. *Significantly different from pretraining (45UT); $^{\Delta}$significantly different from rest; $^{+}$significantly different from 65UT; and $^{\#}$significantly different between resting conditions, $P < 0.05$. From Friedlander et al. (45) and used with permission.

by comparing the fourth with other exercise histogram bars, training increases FFA flux and rate of oxidation (Rox, not shown) for a given REL. After training, for young women the REL task at 65% of posttraining VO_{2max} elicited a 43% greater than that during exercise at 65% of VO_{2max} pretraining (65UT) reflecting a significantly greater capacity for fatty acid mobilization, circulation, and use.

Comparison of the posttraining results for women (Fig. 9A) and men (Fig. 9B) shows gender dimorphism with respect to the effect of training on FFA use during hard (65% VO_{2max}) exercise. Whereas in women FFA Rd is greatest in the REL trial, after training men show suppression of fatty acid disposal during the hard (REL) posttraining trial. This result is significant in several ways. Recalling first that both men and women demonstrate increased FFA Rd and Rox when exercising at a given power output (i.e., the ABT vs. 65UT comparisons) it is to be appreciated that endurance training both increases lipid use (Fig. 9) and decreases dependence on glucose and other CHO-derived fuels (Figs. 7 and 8). Recall also that after training, the power output needed to achieve 65% VO_{2max} is 40% to 50% greater than before training. Seen in this context the rise of FFA Rd and Rox by women during the REL trial is particularly significant as they are better at resisting downregulation of FFA mobilization and use during hard exercise than are men. This gender difference may be related to lesser sympathetic drive that subsequently lessens glycogen degradation rate (48) in women, and also to other factors such as a greater percentage of Type I muscle fibers.

Given complexity of substrate-substrate interactions during exercise, in Figure 10 an attempt is made to reconcile data obtained using D2-glucose, [1-^{13}C]glucose, D5-glycerol, [1-^{13}C]palmitate tracers to describe the fates of FFAs mobilized during exercise and their relative roles as energy substrates during exercise. Figure 10A shows that a significant portion of FFAs mobilized during exercise are returned to storage via reesterification. While significant, the values (approximately 20% during rest and exercise) in young women pale in comparison to the values in postmenopausal women in whom reesterification accounts for more than 50% of fatty acids mobilized (74). As well, Figure 10A is instructive in showing that plasma FFA oxidation does not account for all of total body lipid oxidation. Use of these "other" lipid energy sources including IMTGs, are discussed below. Again, Figure 10B is instructive in showing that plasma FFA and other lipids provide most of the energy used in postprandial humans. However, when exercise starts, CHO-derived energy sources predominate regardles of gender or age (82).

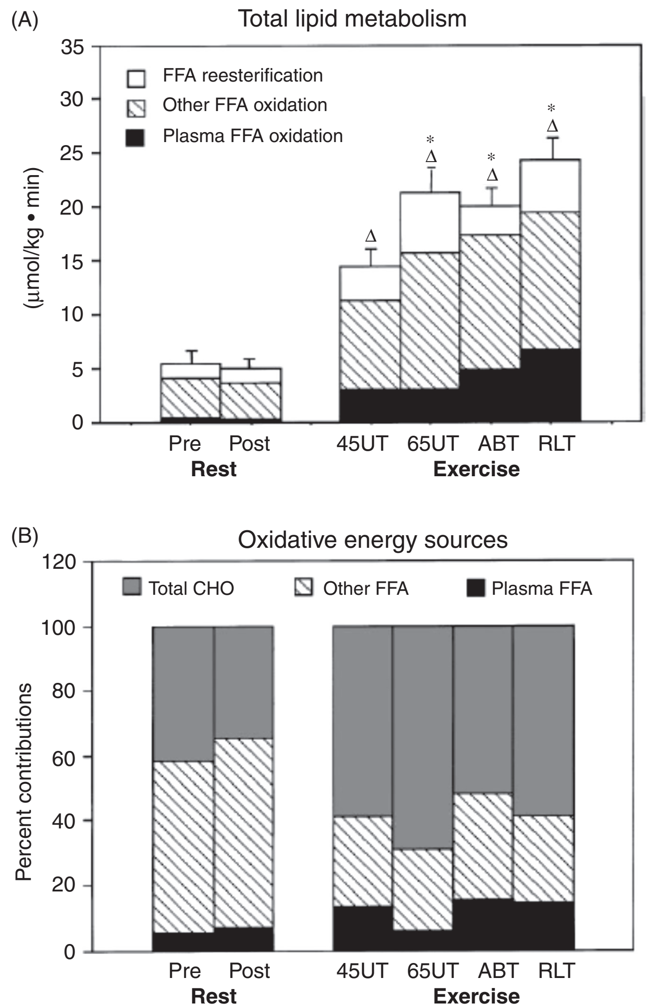

Figure 10 (A) Contributions of different lipid sources to total lipid metabolism during rest and exercise before and after training. SE and statistical symbols are for total lipid metabolism only; $n = 8$, mean ± SE. $^{\Delta}$Significantly different from rest and *significantly different from 45UT ($P < 0.05$). (B) Contributions of energy from different substrate sources during rest and exercise normalized to percent energy expenditure; $n = 8$. From Friedlander et al. (46).

Exercise and exercise training effects on muscle IMTG net use

In the same study providing results of working limb RQ (Fig. 3), IMTG use during exercise was assessed in two ways. First, biopsies taken before and after exercise showed no significant net change (12). Admittedly, however, sampling errors in biopsy studies make interpretation difficult (130). Fortunately, measurements of blood [glycerol] is more sensitive, reliable, and consistent than are biopsy measurements. Most importantly, results of a-v differences in glycerol concentration and muscle IMTG measurements yielded consistent results. Significance of those results is to be appreciated as it is realized that muscle and adipose tissues lack glycerol kinase. Hence, glycerol released from the working limb would be consistent with IMTG mobilization. Regrettably, however, in no condition studied was significant net glycerol release observed. Rather, net limb glycerol release was a feature of postabsorptive rest, whereas in all exercise conditions, whether it is easy (45% VO_{2peak}) or hard (65% VO_{2peak}) exercise, either before or after training, glycerol net release was insignificant (Fig. 11). Moreover, an integrated analysis of Lox during exercise, based on pulmonary RER, working limb RQ, glycerol release (approximately zero during exercise), net FFA uptake (very small during exercise), and insignificant net IMTG content change, inexorably leads to the conclusion that CHO, not lipid energy sources sustain working human skeletal muscle, regardless of training state.

Effects of exercise and exercise training on circulating lipoproteins and lipoprotein exchange across working muscle

In a subsequent study to corroborate and extend previous results (12), determine the availability of lipoproteins and their triglyceride and cholesterol contents, as fuels for

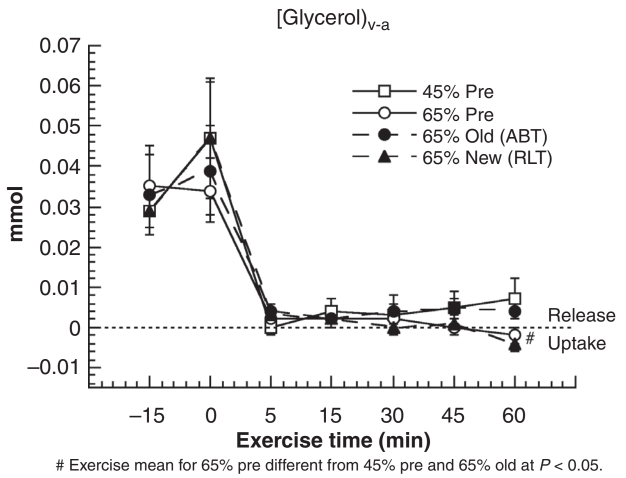

Figure 11 Effect of exercise intensity and training on glycerol venous-arterial difference. Values are mean ± SEM for eight subjects. Little net glycerol release occurs from working limb muscle indicative of insignificant intramuscular triglyceride (IMTG) mobilization during contractions. Symbols represent moderate intensity exercise before endurance training (45% Pre), hard exercise before training (65% Pre), after training the same absolute (ABT) intensity that elicited 65% VO_{2peak} before training (65% Old), and exercise that elicited 65% of the post-training VO_{2peak} [new, relative hard exercise (RLT)]. From Bergman et al. (12) and used with permission.

working human muscle. Accordingly, we studied eight additional men, during rest and exercise, before and after nine weeks of endurance training. Previously, based on working limb RQ (Fig. 3) and net glycerol release (Fig. 11), we assumed insignificant contributions of lipoproteins. Based on recently obtained results (79), the assumption of insignificant muscle lipoprotein uptake and oxidation during exercise in earlier studies (*vide supra*) is justified. Importantly, nine weeks of endurance training favorably affected the postabsorptive circulating lipoprotein patterns. These "long-term" results are in contrast to those in Figure 6 which show no acute effect of exercise, whether the intensity be moderate (45% VO_{2peak}) or hard (65% VO_{2peak}), either before (UT) or after training (T).

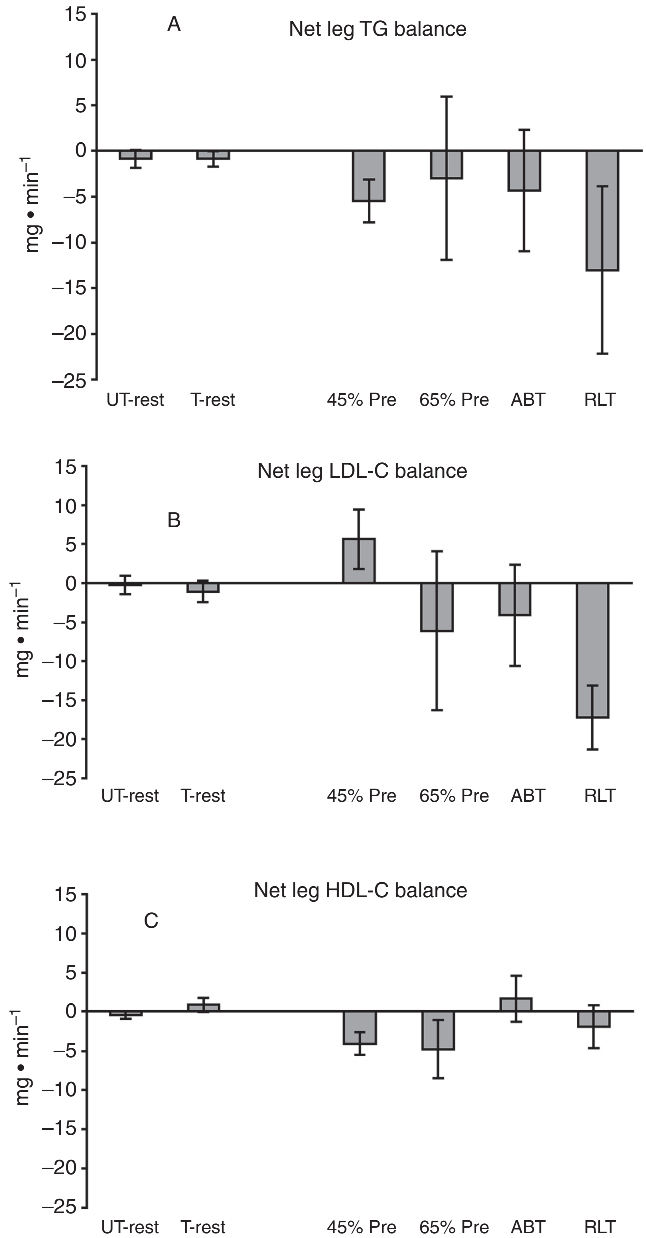

Figure 12 Effects of graded exercise and exercise training on resting and working limb triglyceride (A) and cholesterol LDL-C (B) and HDL-C (C). Exercise and exercise training effects are physiologically insignificant. From Jacobs et al. (79) and used with permission.

Studies of energy substrate use during exercise and recovery from exercise using indirect calorimetry

Investigators have long debated whether the exercise-induced elevation in metabolism should be considered as part of the energy cost of exercise. Classically, the oxygen consumption during the postexercise (O_2 debt) period was determined to assess the extent of energy derived from nonoxidative energy sources during exercise (70, 95). On the contrary, for reasons reviewed elsewhere (51), the period of excess postexercise O_2 consumption (EPOC) has multiple causes, reflecting a general exercise-induced effect on metabolism, and not quantitatively equivalent to ATP produced by lactate formation during exercise. However, the EPOC period possesses utility for assessing total energy expenditure associated with physical activity, especially if a goal is to know the nutritive requirements of physically active individuals (20). To evaluate the hypothesis that physical exercise, even rigorous and prolonged exercise, results in large energy expenditure, but little lipid oxidation, we examined healthy men and women during and after two exercise tasks [89 min @ 45% and 60 min @ 65% of peak rate of oxygen consumption (VO_{2peak})] as well as a time of day-matched resting control trial. Exercise bouts were matched for energy expenditure, the harder exercise task being shorter in duration (60 min), than the easier, but longer task (89 min).

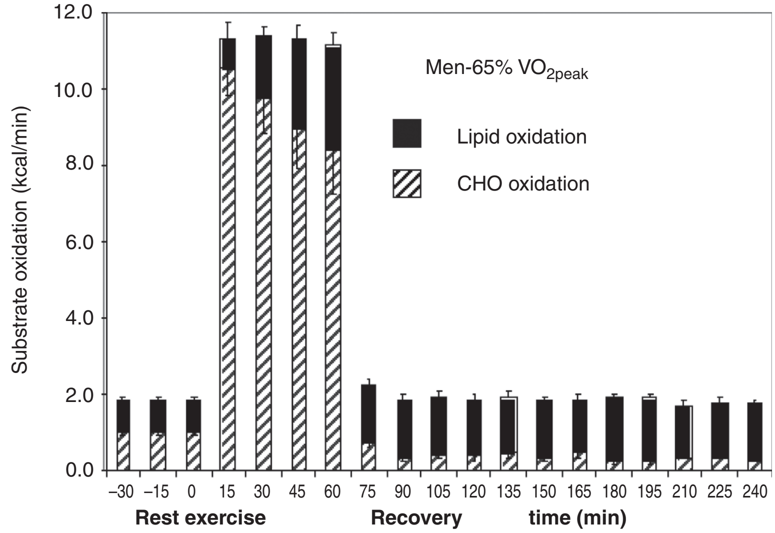

Figure 13 Absolute substrate oxidation rates (mean ± SEM) in men before during and after 60 min of exercise at 65% VO_{2peak}. Lipid and CHO oxidation rates shown by solid and crosshatched bars, respectively. Data show dominance of CHO oxidation during physical activity, and crossover to lipid oxidation during recovery. Data on women are not shown. From Kuo et al. (88) and used with permission.

RERs and VO_2 data were used to calculate lipid and CHO oxidation during recovery. The results showed that men and women switched from predominant CHO oxidation during exercise to lipid oxidation during recovery (88). The switch to predominant lipid oxidation was true even during recovery from the harder (65% VO_{2peak}) trial in which CHO was the major fuel during activity, particularly in men (Fig. 13). The results, subsequently confirmed (63,64), showed the presence of a gender paradox wherein women derive a greater percentage of energy from lipid oxidation during exercise and men depend more on lipid, and less on glucose oxidation during recovery.

Studies of energy substrate use during exercise and recovery from exercise using tracers

Given results of Kuo et al. (88) indicating a shift to reliance on lipid oxidation during postexercise recovery, we studied 10 young lean men and women before and after 90 min @ 45% and 60 min @ 65% VO_{2peak}. To determine exercise and postexercise effects on metabolite fluxes, combinations of D2-glucose, [1-^{13}C]palmitate, and D5-glycerol to determine glucose, FFA and glycerol fluxes, respectively. As well, separate time of day resting trials, and trials using [^{13}C]bicarbonate to evaluate the extent of tracer retention in CO_2 and carbonic acid pools control trials were conducted (62). Results of metabolite tracer studies corrected for [^{13}C]bicarbonate retention (63,64) (Fig. 14) confirm and extend those of Kuo et al. (88) showing that exercise causes increased fat mobilization, flux, and oxidation, and that these exercise-induced effects persist well into recovery.

Fat as a fuel in the absence of others

The preponderance of evidence, on humans (Fig. 4) and other mammals (Fig. 5) is that when energy demand is high, glycolytic flux is high and lipid oxidation is downregulated. Belief in the CHO dependence of athletes is empirically based, and universal with athletes across the endurance spectrum seeking CHO nutrition for energy. For example, human sprint athletes depend heavily on glycogenolysis and glycolysis

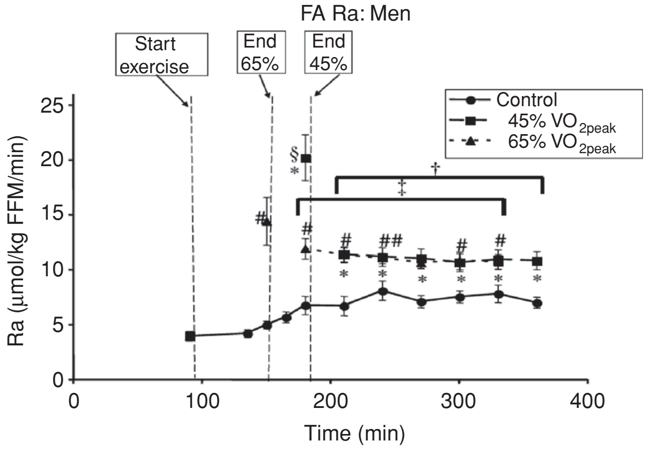

Figure 14 Plasma FFA rate of appearance (Ra) as determined from continuous infusion of [1−^{13}C]palmitate in 10 men studied at rest and during and after two exercise intensities (45% VO_{2peak} for 90 min and 65% VO_{2peak} for 60 min). The same subjects were also studied on a nonexercise day (control) to account for diurnal variations. Plasma FFA Ra rose significantly during exercise, compared to preexercise rest, and remained elevated in 3 h of recovery whether compared to preexercise rest or time of day matched resting control. From Henderson et al. (64) and used with permission.

(Table 1), and as well, endurance athletes, such as ultra-distance runners, bicyclists, and triathletes crave carbohydrate foods both during and after exercise. The bottles carried by amateur and professional cyclists contain dilute aqueous blends of carbohydrates (lactate polymers, glucose, fructose, sucrose, and glucose polymers) (4,81). Few have tried to make a go of endurance cycling with vegetable oil in their bottles.

Despite the habits of successful human athletes, are there examples in nature, athletics or human experience demonstrating that fat feeding promotes lipid oxidation that in turn, engenders exercise endurance? In the animal kingdom, extraordinary access and use of body lipid stores is demonstrated by migratory birds (106). Among mammals, extraordinary running endurance is demonstrated by sled dogs fed high-fat diets (58). In human experience, individuals and populations have adapted to very different diets, ranging from running successes on grain-based diets used by Kenyan Olympians (104), to dependence on marine mammals and fish by humans native to northern climates and arctic explorers of European descent (117). Tales of arctic explorers who live on fatty meats alone are legendary; the ability of early 20th century English explorers to toil by sledge hauling and exist on fatty meats when necessary borders on the unimaginable (32).

Knowing metabolic plasticity as well as having read the accounts of Lieutenant Frederick Schwatka, Aspley Cherry-Garrard and others, Phinney and colleagues set out to evaluate whether members of a professional bicycling team could adapt to a low CHO, ketotic diet (105). Realizing that adaptation to a ketotic diet takes several weeks, Phinney studied athletes before and after four weeks of a high fat, high protein, and low CHO diet. After the ketotic diet, athletes were able to maintain capacities for VO_{2max} and submaximal endurance at 62% to 64% VO_{2max}. However, the performance of athletes during hill climbing was diminished (S.D. Phinney, personal communication). The latter effect was likely attributable to decreased muscle glycogen content.

More recently Hawley and Burke (60) reviewed results of their efforts and those of others who have explored the possibility of increasing the stress on muscle during training by high-fat and low-CHO feeding. As they point out, preparation for competition involves some or all of the following: training in the morning after an overnight fast, training involving weight loss, twice a day training, and prolonged training sessions without taking nutrition. All of these maneuvers cause athletes to work while becoming glycogen depleted with glycemia being supported by hepatic gluconeogenesis. Also as they point out, CHO-deprivation significantly decreases power output in training, but the relatively greater training stress may increase factors that stimulate muscle GLUT4 protein expression and mitochondrial biogenesis. The purported mechanisms are postulated to involve activation of AMP-activated protein kinase (AMPK), the expression of transcription factors, specifically nuclear respiratory factors-1 and -2 (NRF-1 and NRF-2), and the expression of peroxisome proliferator-activated receptor gamma coactivator-1 alpha (PGC-1α), as well as the nutrient-sensitive signaling molecule p38 mitogen-activated protein kinase (MAPK) that phosphorylates and activates PGC-1α.

Realizing that training on low-CHO diets might stimulate muscle adaptations, but that CHO is necessary for good performance, some investigators, for example, Havemann et al. (59) trained cyclists on a high fat diet (68% of energy intake) for six days, followed by one day of a high-CHO, glycogen loading diet. This treatment had the effect of lowering RER (increasing lipid oxidation) during a 100 km bicycle time trial, but performance was unaffected. In contrast, performance in a 1 km sprint was negatively affected by the high-fat diet followed by one day of CHO loading. Other than measuring VO_2 and RER, there is a dearth of efforts using isotope tracers to know the effects of fat feeding followed by a period of rest, with and without CHO loading on energy substrate partitioning and criterion exercise performance. In this context, what do the results show?

Once again, results of the fat-feeding maneuvers demonstrate the dedication of athletes to their sport and the equally remarkable ability of muscles and related organ systems, such as liver to provide energy substrates for exercise. The human body can adapt to a ketotic diet, and this may be appropriate for an arctic explorer, but short- or long-term benefits of fat feeding on exercise performance are not apparent. To the contrary, those who have studied the acute effects recommend against high-fat feeding as a means to enhance exercise performance (25, 26, 60). As well, from the standpoint of long-term health, high-fat feeding is counterproductive (20).

Steady Rate Ergometry and Problems in Computing the Efficiency of Human Locomotion

Realizations about the integrated functions of muscle energy transduction systems and the use of energy substrates to operate the apparatus for excitation-contraction coupling lead to concepts of parsing the totality of energy conversions into two components: the phosphorylative coupling and mechanical coupling efficiencies. These discrete efficiencies can be measured on isolated systems *ex vivo* (*vide supra*). Alternatively, phosphorylative coupling and mechanical coupling efficiencies can be estimated on intact working individuals when one or the other can be determined. Realizations about limitations in our ability to measure components of coupling efficiencies during nonsteady states has placed emphasis on making determinations during steady-rate conditions and then employ mechanical maneuvers and computational modalities to devolve the separate coupling efficiencies and their interactions. Measurements of the efficiency and economy of human locomotion suggest maximal efficiencies of 25% to 35% in leg cycling (50) and similar, if not slightly higher efficiencies for normal gait walking (36). Differences in efficiencies measured during cycling versus walking may relate to the ability to transfer the kinetic energy of one step to the next in walking whereas there is little kinetic energy transfer within or between limb muscles during leg cycling. Other modes of transport, such as slow walking (30, 109) and swimming (40, 76) yield lesser values.

The baseline hurdle

Pulmonary and muscle metabolism rises in response to the challenge of doing muscle work (Fig. 1), but how much of the rise is attributable to muscle work is a longstanding question. The assumption that basal and other housekeeping processes continued during exercise led to several approaches to adjust (correct by subtraction) for the background energy costs unrelated to muscle work; this is the matter of the so called "baseline correction" (50, 51, 120).

Net efficiency

One approach to correcting for basal and housekeeping functions during exercise is to subtract a constant quantity, such as the preexercise resting metabolic rate from the metabolic rate observed during exercise. "Net" efficiency is then:

$$\textbf{Ef}\ (\text{Net})\ (\%) = (100)\ \text{external power (kcal)}/\ \text{exercise } VO_2\text{-resting}VO_2\text{(kcal)}.$$

Using resting metabolic rate as a baseline correction has the effect of depressing the calculated efficiency at low power outputs, but as power output rises subtracting the resting value from exercise energy expenditure has less of an effect, and the calculated net efficiency rises. However, with reference to Figure 15, the no (0) work y-intercept differs from the resting metabolic rate ($\approx$ 1 kcal/min) as well as the "zero power," unloaded cycling rate that is shown at the y-intercept.

Gross efficiency

Another approach to correcting for basal and housekeeping functions during exercise is to make no correction for the metabolic power associated with unloaded exercise, but zero is a constant and has similar effects as does the resting metabolic rate subtraction in determining net efficiency. "Gross" efficiency is then:

$$\textbf{Ef}\ (\text{Gross})\ (\%) = (100)\ \text{external power (kcal)}/\ \text{exercise } VO_2\ \text{(kcal)}.$$

Yet again, with reference to Figure 15, the no (0) work y-intercept differs from zero, and the computed gross efficiency rises as exercise power output rises (Fig. 22B, *vide infra*). This apparent rise in exercise efficiency occurs while the slope of rise of energy expenditure is constant or increasing (Fig. 15). Hence, both net and gross efficiency calculations suffer from the same two artifacts.

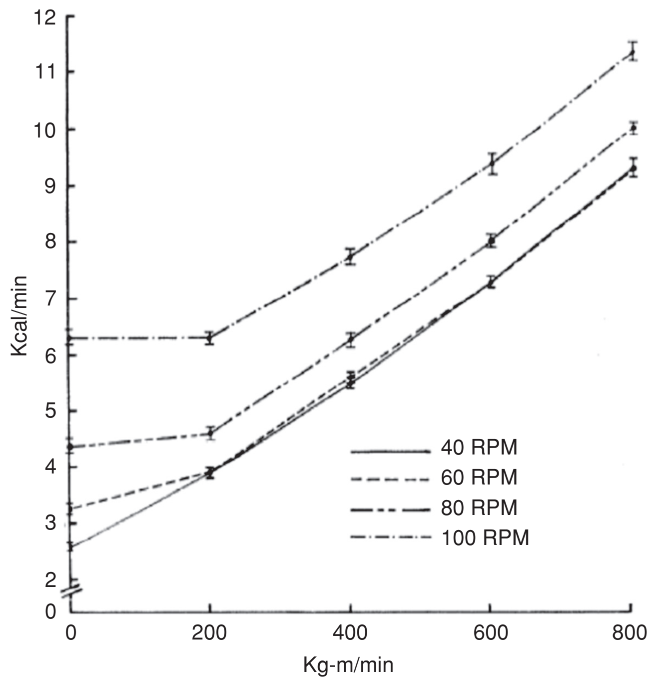

Figure 15 Effect of work rat and speed of movement on a leg cycle ergometer (mean ± SE) of 12 young males during steady-rate exercise. Caloric values determined from VO_2 and RER. The essentially linear relationship with perhaps a slight exponential rise in caloric output at higher work rates dictates either constant or decreasing efficiency. Because the caloric cost of each exercise power output increases with increments in pedaling speed, decreasing efficiency with increasing speed is indicated. Note that the y-intercepts measured during unloaded cycling deviate from rest ($\approx$ 1 kcal/min) and significantly from linearity for all but the slowest cycling cadence making "gross," "net," and "work" efficiency calculations invalid, and indicating use of the "delta" method of efficiency calculation. From Gaesser and Brooks (50) and used with permission.

Delta (Δ), instantaneous and work efficiencies

As opposed to fixed baseline corrections, there are three other approaches to calculating muscular efficiency. In common, these approaches attempt to use an exercise baseline to correct for basal, housekeeping and other metabolic costs extraneous to delivering muscle power. The first of these is "work efficiency", in which unloaded movement, such as free wheel cycling is conducted and the values used to correct the metabolic cost when a load is applied. Although insightful and simple in concept, even for bicycling in practice it is difficult to measure unloaded cycling. For example, as seen in the y-intercepts in Figure 15, frictional loads and eddy currents in cycle ergometers cause the y-intercept determined during unloaded cycling to be off the regression lines for most conditions studied. Hence, for leg cycling because of problems in controlling the ergometer at zero power, and also likely because of variations in human movement patterns during unloaded cycling, it has proven difficult to obtain a "zero" work exercise baseline. And, to this writer's knowledge, while partial unweighting by suspension has been attempted for walking subjects (56), and while other forms of pneumatic and aquatic suspension are used in rehabilitation medicine, physical suspension with counterweights, springs and other means of unweighting have not been successfully applied to the computation of work efficiency during walking.

Delta (Δ) efficiency

The (Δ) approach to calculating muscular efficiency utilizes a floating baseline that varies as the metabolic response to graded exercises changes. In this way efficiencies can be computed all along a power output curve such as seen in Figure 15. "Delta" efficiency is then:

$$\textbf{Ef}\ (\Delta)\ (\%) = (100)\ \text{change in external power (kcal)}/\ \text{change exercise}\ VO_2\ \text{(kcal)}.$$

Or, as shown in Figures 2 and 15, inverse of the regression of the caloric equivalent of pulmonary VO_2 on the caloric equivalent of external power output during leg cycling for power outputs > 200 kg-m/min.

A variation on Δ efficiency is "instantaneous efficiency" which is computed as the reciprocal of the first derivative of the equation describing the relationship between energy expenditure and work rate (36). This approach has the effect of smoothing the curve, but some information may be lost at ends of the power output curve. Similarities and differences between delta and instantaneous efficiencies are shown below (Fig. 22).

Human muscular efficiency In cycling and walking

In a metabolic state when power output is submaximal and steady such that the rate of oxygen uptake (VO_2), pulmonary respiratory gas exchange ratio (RER) and blood [lactate] are constant (steady rate determinations for constant VCO_2/VO_2 and [blood] are typically made over periods of 3-10 min), the "energy cost" of exercise can be calculated from standard tables developed by Zuntz and Schumburg (137) developed at the end of the 19th century and modified by Lusk early in the 20th century (62), or by a theoretical-thermodynamic approach (50,132). These computations assume that energy is derived from the oxidation of carbohydrate and lipid energy sources. From the data in Figure 15 an example of the use of energy expenditure data to compute the efficiency of leg ergometer cycling at low pedal cadence (40 rpm) is given in Figure 16, where the effect of increasing power output is shown to decrease efficiency. The similar computation for high cadence (100 rpm) cycling (Fig. 17), again shows a decline in computed efficiency as power output increases. Efficiency values are slightly higher for cycling at 100 vs 40 rpm, but are in the same range.

Some of the unresolved problems in evaluating the effects of changes in speed and power on the exercise efficiencies shown in Figures 16 and 17 can be seen by close

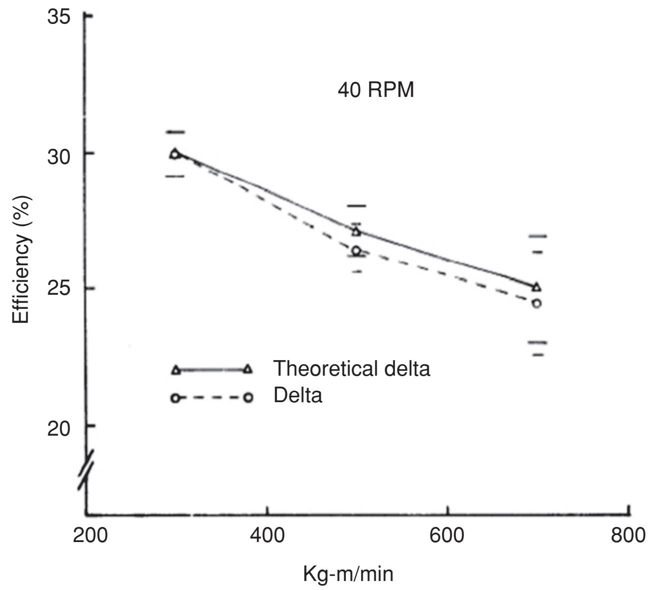

Figure 16 The effect of work rate on delta (Δ) efficiency (mean ± SE) for 12 young men pedaling at a slow cadence (40 RPM) on a leg cycle ergometer. In contrast to other (gross, net, and work) modes of calculation, the data (From Fig. 15) demonstrates decreasing efficiency with increments in power output. From Gaesser and Brooks (50) and used with permission.

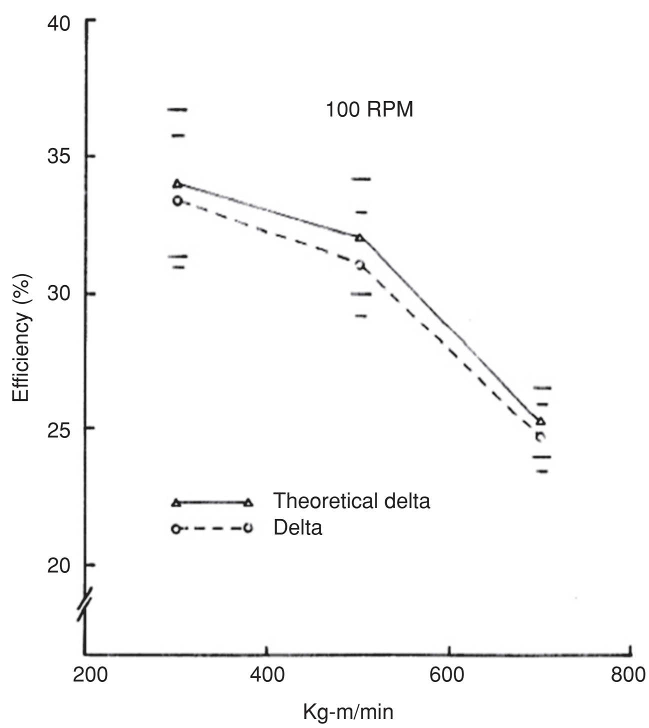

Figure 17 The effect of work rate on delta (Δ) efficiency (mean ± SE) for 12 young men pedaling at a rapid cadence (100 RPM) on a leg cycle ergometer, calculations based on data in Figure 15. As with slow cadence pedaling (Fig. 16), the results demonstrate decreasing efficiency with increments in power output when cycling at a fast cadence. Note that the range of computed efficiencies (25%-35%) is similar for low and high cadence pedaling. From Gaesser and Brooks (50) and used with permission.

inspection of Figure 15 which gives the data upon which those efficiencies were calculated. The literature contains extensive commentary on those issues (36, 50, 51, 80, 120, 132). Beyond the assumption of no change in basal energy expenditure is that of the changing y-intercept. Specifically, in comparing cycling cadences of 40 and 100 rpm the caloric expenditure for high cadence cycling is clearly greater than when the same external power output is accomplished during cycling at lower cranking rates (Fig. 15). Recognition that energy cost represented by the y-intercepts in Figure 15 likely includes energy associated with internal work accelerating and decelerating the legs and increases in the metabolic costs of ion pumping associated with more frequent, although less forceful contractions. But, who can claim that internal work that gives rise to increased energy expenditure is not truly "work" as is measured in a Newtonian sense? Therefore, in the example shown (Fig. 17), the mode of computation attempts to account for both the effects of increased power and greater internal energy cost of cycling at high, 100 rpm by using as the baseline delta in metabolism the energy cost measured during 200 kg-m/min less power and cycling at 20 lower rpm.

In aggregate, comparing Figures 15 and 17 illustrates how investigators need to exercise judgment when interpreting and applying steady-rate efficiency calculations. While the computed efficiencies for cycling at 40 and 100 rpm (Figs. 16 and 17), and the actual metabolic energy cost for cycling at 100 rpm is greater than for cycling at 40 rpm (Fig. 15), no competitive or recreational bicyclist would choose pedaling at 40 rpm over pedaling at 80 to 100 pm to maintain a given ground speed. This is because the greater force required to push the pedals at slow cranking rate requires greater force and recruitment of fast-fatiguable fibers (55), a decision that would result in poor, fatigue-limited performance.

Values given in Figure 15 and corresponding efficiency calculations (Figs 16 and 17) are for leg cycling, a mode of laboratory ergometry that is ideal for investigators and which is amenable to human locomotary patterns. In contrast, walking is a movement pattern for which humans are evolved, but which presents problems for investigators interested in relating the energy cost of movement to external physical work done. For humans walking or running on a horizontal treadmill, no external work is accomplished. Accordingly, to know external work accomplished, investigators have used inclined treadmill exercise where work of lifting the body can be determined (118). As well, to know the work accomplished in forward locomotion, investigators have used wind tunnel exercise (108), or a horizontal impeding force (92), as originally applied to measuring horsepower (86). As well, engineers have calculated the work done in accelerating and decelerating limbs during walking (31, 109, 110). One example of the "vertical" work of treadmill climbing is depicted in Figure 18 (36). In that experiment, treadmill speed and gradient were manipulated, and knowing subject mass the efficiency of vertical work could be estimated. In the example shown, the efficiency of gradient climbing (Fig. 18) decreases

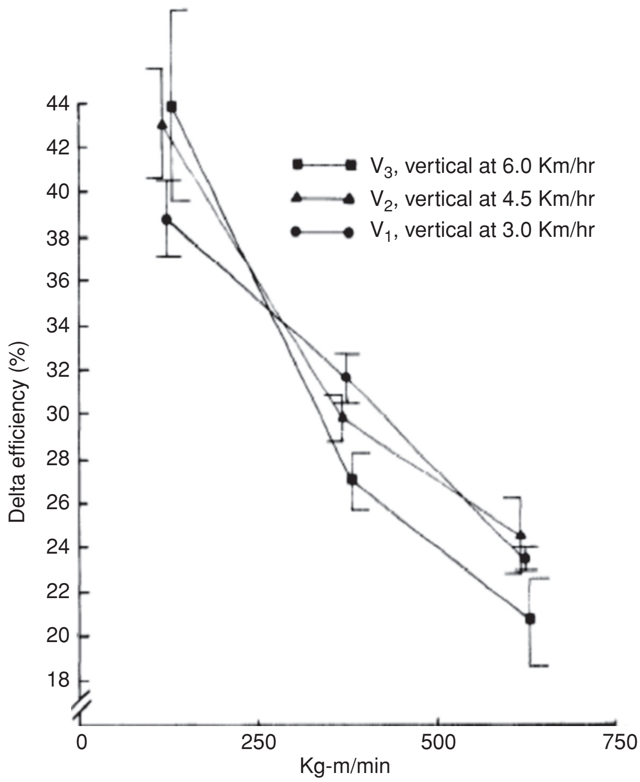

Figure 18 Effects of work rate on delta (Δ) efficiency (mean ± SE) for nine young men during treadmill gradient walking at 3.0, 4.5, and 6.0 km/h. Results gradient (vertical) work as well as horizontal work against an impeding force demonstrate decreasing efficiency with increments in muscle power output. As with results in [Figs. (16, (17)], computations based on caloric values determined from VO_2 and RER during steady-rate, submaximal exercise. From Donovan and Brooks (36) and used with permission.

with increasing exercise power outputs, as it does in leg cycle ergometry (Figs. 16 and 17).

Classic Compared to Theoretical-Thermodynamic Approaches to Estimating Human Exercise Efficiency *In Vivo*

The presence of parallel and superimposable plots lends confidence in the underlying assumptions

Figures 16 and 17 show two lines each, one line representing delta efficiency computed by the "classic" method, and the other computed by "theoretical-thermodynamic" method. As explained above (50, 80, 120, 132), delta efficiency was based on the regression of metabolic energy expenditure (determined by indirect calorimetry) on external power output. In the calculation of delta efficiency, ergometer power output is changed and the corresponding changes in oxygen consumption and carbon dioxide production are observed and used to calculate the change in metabolic energy expenditure using the tables of Zuntz and Schumburg (137), as modified by Lusk (93). The change in metabolic energy expenditure is then compared to the change in external power output, and the computation becomes the inverse of the slope of metabolic power output. As illustrated in Figures 16 and 17, a leg cycle ergometer can be used to determine the metabolic responses to ordered changes in mechanical power output over a wide range. Delta efficiency calculations such as those in Figures 16 and 17 have the advantage of being sensitive to changes in RER, and, therefore, energy expenditure as external power output rises the relative use of CHO oxidation increases.

The "theoretical delta" efficiencies were based on results of measurements made using assumptions made *ex vivo*. Specifically, the assumptions were that the P/O was 3, that the free energy of ATP (ΔG') for ATP was –11 kcal/mol, and that the ATP yield from glucose oxidation was 38 mol ATP/mol glucose. Accordingly using data from Lehninger (91), the theoretical-thermodynamic calculations for glucose oxidation were obtained: efficiency = [(38 ATP/glucose) (−11 kcal/mol ATP) = −418 kcal captured as ATP]. For, glucose with an enthalpy (heat of combustion, $\Delta G' = -686$ kcal/mol glucose), the theoretical coupling efficiency of oxidative phosphorylation related to glucose oxidation is then (418 kcal/mol ÷ 686 kcal/mol = 0.61, or) 61%. As shown clearly in the figures, the two efficiencies assume mechanical coupling efficiencies of 50% and produce overlapping results. These results obtained on healthy exercising adult humans are the strongest available to support validity of the P/O and ΔG' results obtained on systems *ex vivo*.

For glucose degradation, there are two segments in catabolism, glycolysis and subsequent oxidation of the products of glycolysis (lactate anions). Because the individual enthalpies of the two segments are known, the separate efficiencies of the two segments can be estimated. For the conversion of glucose to lactate, $\Delta G' = -38$ kcal/mol. From the energy released in glycolysis, the net ATP yield is 2 mol lactate/mol glucose, or [(2 ATP) (−11 kcal/mol) = 22 kcal/mol]. The efficiency then = [(−22 kcal/mol) / (−38 kcal/mol) = 0.58 or] 58%.

In the second segment, for the oxidation of lactate $\Delta G' = -326$ kcal/mol and the ATP yield is 17 mol ATP/mol glucose. Accordingly, energy yield captured as ATP = [(17 ATP/glucose) (−11 kcal/mol glucose = 187 kcal]. Hence, for lactate oxidation the coupling efficiency = 187 kcal/mol ÷ 326 kcal/mol = 0.57, or] 57%. In reality then, phosphorylative coupling efficiency for glucose, as well as for the glycolytic and mitochondrial segments are similarly efficient regardless if calculated from traditional indirect calorimetry or using assumptions derived from data obtained *ex vivo*. The small differences obtained are likely attributable to the assumptions used. For example, the P/O ratio of 3.0 is a theoretical maximum for NADH-linked substrates, whereas the P/O for FADH2-linked substrates is 2 such that a P/O ratio of 2.83 might be appropriate *in vivo*, as might a ΔG' of –10.5 kcal/mol ATP and an ATP yield of 36 mol ATP/mol glucose.

The calculation of theoretical delta efficiencies as in Figures 16 and 17 provide invaluable insight into how to compare whole body respiratory data with those obtained on isolated fragments of the mitochondrial reticulum. Seeing that efficiencies computed by classical and theoretical means, and that both methods respond to increases in speed and power output is, in fact, remarkable. In the view of this writer, the close correspondence of results is the best evidence that the use in theoretical calculations (i.e., P/O for NADH-linked substrates, P/O = 2 for FADH2-linked substrates, ATP yield from glucose = 36-38, and $\Delta G_{ATP} = -10.5$ kcal/mol) are either correct, or very close to being correct.

The values cited above and used in the calculations presented here are taken from Lehninger (91) and other classical sources. In fairness, it needs to be said that in the past (71,94), as well as more recently (18) investigators have reported lesser values, for example, 30 to 32 ATP/glucose and P/Os of 2.5 for NADH-linked and 1.5 for FADH2-linked substrates, respectively. However, substrate-level phosphorylation of ADP occurs in glycolysis, and *in vivo* most glycolytic carbon flux is from glycogen increasing the P/O from glycolysis 1.0/G6P. As well, variability in measuring the P/O of various substrates in isolated mitochondria can be understood from the perspective of new concepts of the mitochondrial reticulum (85) and mitochondrial dynamics (39).

In muscle, and every other tissue examined, the respiratory apparatus is arranged as a network, a mitochondrial reticulum (85). Moreover, the reticulum is not static, but turns over continuously at rates depending on several factors, mostly related to energy state (39). Accordingly, what we obtain when preparing mitochondria for examination *ex vivo* are, in fact, fragments of the mitochondrial reticulum. With that view, it is unrealistic to expect the fragments to work as well as they do *in vivo*. Accordingly, the data from leg cycle ergometry are best assessing mitochondrial efficiency *in vivo*. Simply, and energy yields of 20 to 32 ATP/glucose and P/Os of 1.5 to 2.5 would not allow classical and theoretical-thermodynamic calculation of human muscle efficiency (Figs. 16 and 17) to be as close as they are. As well, because delta efficiency is determined from slopes of changes in metabolic versus mechanical power outputs, small changes in P/O and ATP yield per unit substrate are compensated for in the mode of calculation.

The Steady VO_2 Rate and the Assumption of Minimal ATP Production via Anaerobic Glycolysis

Steady-state data obtained in the last several minutes of 6 to 8 min of constant rate submaximal exercise the assumption is that VO_2 and VCO_2 measurements accurately represent ATP turnover during the measurement period. This means constant levels of muscle adenosine triphosphate ATP, PCr, and lactate. These assumptions about the phosphagens ATP and PCr are likely justified for steady-rate, submaximal exercise (77). This assumption acknowledges that while the ATP and PCr pools constantly turn over; the rates of hydrolysis to support cell work are matched by the energy captured in oxidative phosphorylation. However, the question arises, do constant muscle and blood lactate levels mean the absence of significant "anaerobic" ATP production in glycolysis? The answer is almost certainly "yes."

Recent knowledge that lactate is formed continuously, even in resting individuals and that perturbations such as carbohydrate eating (44) and physical exercise (15, 96, 121, 122) increase lactate production under fully aerobic conditions (111). At issue then is whether the elevated lactate production during muscle exercise results in significant ATP production not accounted for in the measured oxygen consumption. Because most (75%-80%) lactate is disposed of within working skeletal muscle (15, 120), heart (52), and brain (127), and most of the remainder converted to glucose (14, 47, 48) is subsequently oxidized, measures of tissue and pulmonary oxygen uptake account for glycolytically produced ATP.

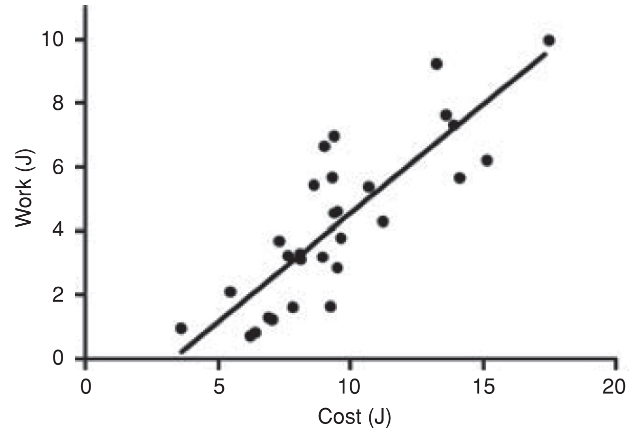

Figure 19 Relationship between external work and metabolic cost of human interosseous (hand) muscle in which metabolic cost was determined by ^{31}P-MRS. The least squares linear regression of data yields a very high, 68% mechanical coupling efficiency (i.e., external work from ATP hydrolysis) [Work output (J) = 0.68 ± 0.09 ATP cost (J) − 2.2 ± 0.9 (J)]. Assuming a phosphorylative coupling efficiency of 50%, overall Δ efficiency approximates 34%. Used with permission. From Jubrias et al. (83) and used with permission.

Real-time Measurements of the Energetics of Working Human Muscle

The assessment of working muscle energetics by means of a-v difference and blood flow measurements such as those of Poole et al. (107) and others (49) provide extra muscular measurements of the energetics of contraction. Recognizing those limitations, at the University of Washington Martin Kushmerick and colleagues (including Kevin Conley and Sharon Jubrias) and others have developed highly sophisticated techniques of phosphorous MRS and muscle ergometry to parse the components of contraction energetics within working human muscle (83, 89). ^{31}P-MRS provides measurements of intramuscular phosphate (Pi), PCr, and ATP. From the chemical equilibrium of creatine kinase, the levels of ADP can be determined; and, assuming a value for the buffering capacity of muscle and chemical shift (splitting) in the Pi peak, muscle hydrogen ion (H^+) concentration is calculated. To go from static measurements of ATP, PCr, and [H^+] a period of ischemia is imposed followed by return to free flow conditions all the while MRS spectra are acquired. Confinement to the bore of NMR magnets and the necessity to use small, nonferrous ergometers obviates measurements on whole-body exercises such as cycling, walking, running, or swimming. Still, accurate and discrete measurements can be made on small muscle groups made to contract in a confined space. As is always the case, the method requires numerous assumptions that the investigators have systematically evaluated. Of those, assumption about the creatine kinase equilibrium, free energy of ATP hydrolysis, P/O ratio and magnitude and constancy of muscle buffering capacity are certain. Other assumptions such as that the chemical shift in the Pi peak is due to anaerobic glycolysis, and glycerol phosphate shuttle explains cytosol to mitochondrial redox balance, are accepted by most biochemists. However, for physiologists it is certain that, relative to whole-body exercise, during small muscle exercise there are surfeits in capacities for cardiac output and muscle blood flow during the postischemic, hyperemic period. More importantly for them is the assumption that the rate of PCr restoration postischemia equals the ATP flux during free flow exercise. Important also is that the biomechanics of small muscle movements in the magnet bore differ from large muscle exercise. And, perhaps most difficult for the method to overcome is the assumption that glycolysis makes lactic acid, the investigators assuming a 1:1 stoichiometry between net lactate production and H^+ accumulation, an assumption that has been hotly disputed as the terminal step in glycolysis catalyzed by lactate dehydrogenase (LDH) consumes protons and produces lactate$^-$ (114). Nonetheless, phosphorus NRS offers a unique means to noninvasively and continuously determine the ATP flux during human exercise.

One such example of the types of data that can be obtained is shown in Jubrias et al. (83) (Fig. 19). Knowing the ATP flux and with a reasonable estimate of the free energy of hydrolysis for a working human first interosseous dorsal (FID) hand muscle, the investigators could calculate a value for efficiency as the ratio of physical work accomplished (from ergometry) to the metabolic cost (from the ATP) flux. Hence, for the interosseous muscle, the mechanical coupling efficiency was 68% (Fig. 19). This is a probably the highest value reported for any mammalian muscle. Then, assuming an efficiency of 60% for phosphorylative coupling, overall efficiency approximates 41%. This value is significantly (~5%) higher than for human leg cycling (Figs. 16 and 17), but similar to that for slow walking (Fig. 18). Seemingly then, for slow movements, the efficiency of human muscle work can be as high as 40%.

Human Muscle Energetics in the Nonsteady State

Considering the difficulties in estimation human muscle efficiency in the metabolic steady state, the difficulties encountered by those wishing to know the efficiency of short-term, high intensity exercise are enormous. Knowing the efficiency of human muscle during intense exertion has been of interest since the early 1920's when Krogh and Lindhard (87) and Hill, Long, and Lupton (67-70), and Smith (118) could reliably measure oxygen consumption in exercising humans. In particular, Smith was focused on steady-state human energetics, but others recognized the need to estimate energy costs from nonsteady-state exercise. For those purposes, the seminal investigators devised "O_2 deficit" and "O_2 debt" methodologies. Regrettably, for numerous reasons reviewed elsewhere (51), those indirect measures cannot reliably be used to estimate the nonoxidative energy turnover during exercise. How then to estimate the energetics of short-term, high power output exercise?

Using an impressive array of technologies including direct and indirect calorimetry, muscle biopsies, and measurements of tissue metabolite exchange, Bangsbo and colleagues (5) reported ATP flux rates on men during 3-min intense bouts of leg kicking exercise. As explained by them, leg kicking was chosen to emphasize quadriceps exercise, a muscle whose mass could be estimated, that could be biopsied and whose metabolite content (product of concentration and blood flow) could be measured. The rate of muscle oxygen consumption plus an estimation of 50% of myoglobin desaturation gave a value for "aerobic metabolism." Anaerobic metabolism was estimated from the decrements of ATP and PCr, muscle lactate accumulation and net release. Oxidative disposal of lactate was covered in the limb VO_2 measurement. Moreover, they investigated the potential effects of energy provided by "other" energy sources such as pyruvate and alanine release and accumulation, lactate uptake by inactive tissues in the limb. Quantitative estimates of myokinase activity were assumed to be minor and not attempted. Their summed estimates of ATP use in two 2-min exercise bouts separated by 6 min (Fig. 20) show the predominant role of oxidative metabolism beyond 15 s of the 3-min effort; estimates for periods <15 s are uncertain and not shown. PCr pool depletion provides approximately 6% of the energy derived. Estimates for the role of lactate in energy production from net release and accumulation show a 25% contribution. The nonsteady state and absence of a lactate tracer precludes an estimate of the role of lactate disposal via oxidation, that quantity being included in the estimate of aerobic metabolism.

Given that they could estimate the ATP flux and because they knew the external power output the investigators could calculate a "mechanical efficiency" as the ratio of external power output to energy (ATP) expenditure; that efficiency computation (Fig. 21) corresponded to "gross efficiency" as defined by us (50) and others (80). The computed values are not comparable to the delta efficiency values as reported

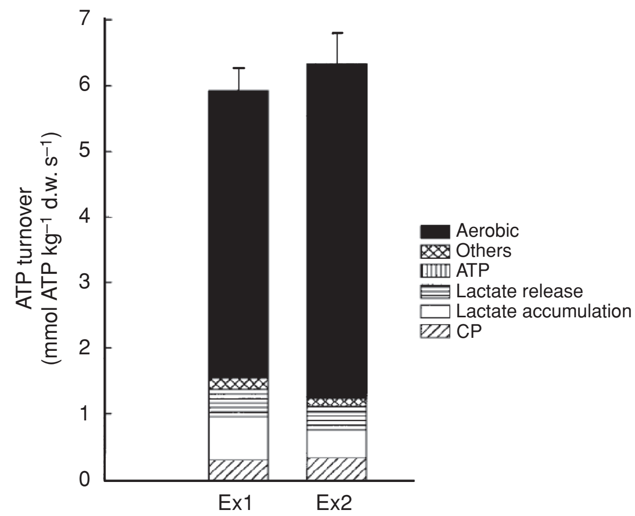

Figure 20 Rate of muscle ATP turnover (mmol ATP/kg dry wt/s) during 0-5, 5-15, and 15-180 s of two bouts of intense knee extensor exercise (EX1) and (EX2) separated by 3 min of rest. ATP turnover estimated as the sum of muscle anaerobic energy production determined as energy release related to utilization of CP (hatched part of bar), net lactate production determined as the sum of accumulation in muscle (open bar) and release to the blood (horizontally lined bar), net ATP utilization (vertically lined bar), others sources, and aerobic energy production (filled bar), determined from muscle oxygen uptake and estimated utilization of oxygen from myoglobin. Values are means ± SE. Modified from Figure 4 in Bangsbo et al. (5) and used with permission.

in Figures 12-16, but the investigators undertook a unique computation that was to estimate efficiency from the ratio of physical work accomplished to the heat liberated during exercise. Symbols in the figure indicate different efficiencies at the beginning and end of the 3-min trial, but no differences

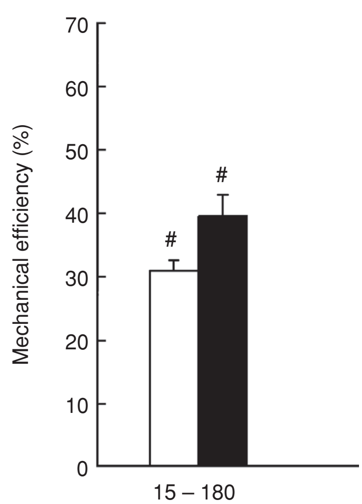

Figure 21 Muscle mechanical efficiency, determined as work per total energy production during the interval between 15-180 s of an intense knee extensor exercise, in which total energy production was determined from metabolic measurements (open bars) and as the sum of total heat production and work performed. Values are means ± SE. #significantly ($P < 0.05$) different from values determined during the first 15 s of exercise. Values are means ± SE. Modified from Figure 6 in Bangsbo et al. (5) and used with permission.

between efficiencies determined by direct calorimetry and ATP turnover.

Muscle Fiber Type and Energetics

Thermopile studies on isolated rat soleus, comprised mostly of slow-red fibers (53), and extensor digitorum longus, comprised of mostly of fast-white fibers (131) provided support for the idea that slow and fast muscles are evolved for different purposes, with slow fibers adapted for maintaining tension, as in postural control, whereas fast muscles are adapted for rapid limb movement (33, 50). In this context, it is notable that some investigators (3) argue that fast fibers are adapted to function at higher speeds and greater power outputs, and are perhaps more efficient working under those circumstances than are slow-twitch fibers. These data on *ex vivo* systems appear to be relevant to the human condition.

Regardless of the data from isolated frog and mammalian muscles, studied in thermopiles and other devices at low temperatures *ex vivo*, and the disagreements around methodology and interpretation of such data (3), it is of interest to know the applications to understanding the bioenergetics of freely moving humans. It is for certain that there exists heterogeneity of muscle fiber type in human populations (55), but little attention has been paid to the effects of muscle fiber type on muscle energetics. One of the few attempts was that of Stuart and colleagues (123) who conducted leg cycle ergometry on track and field sprinter and long-distance runner teammates at a major US university. These sprinters and endurance athletes were selected to represent individuals possessing fast- and slow-twitch fiber types, respectively, and leg cycling was used for experimental purposes as well as to provide a neutral exercise modality that neither group had trained extensively on. An assumption inherent in using runners to study leg cycling economy and efficiency is that both groups were equally naïve to cycling. That assumption seems justified as in a previous study on highly experienced compared to recreationally experienced college males produced superimposable curves of the regression of metabolic rate on external power output (101). In that context the results of Stuart et al. (Fig. 19), support studies on isolated fibers indicating greater efficiency of muscle contraction for individuals expressing predominantly slow-twitch muscle fiber types. In Figure 22A (their Fig. 3) the energy cost of performing identical exercise power outputs is higher in sprinters than in distance runners. Greater energy cost, and hence lesser efficiency of the fast-twitch predominant sprinters is suggested. Accordingly, whether by gross efficiency (Fig. 19B, their Fig. 2), or delta (Fig. 19C, their Fig. 1) efficiency computation, at low to moderate leg cycling power outputs distance runners were more efficient. As well, Stuart et al. (115) also calculated instantaneous efficiency over the range of observed energy expenditures (34). That computation (Fig. 22D, their Fig. 4) had the effect of smoothing the delta work efficiency curves (Fig. 19C). The results are similar to the extent of showing similar efficiencies at the higher (140 W) exercise power output studied. Assuming that the power output studied (50 kJ/min, 12 kcal/min) represented steady-state submaximal exercise for both sets of athletes, the results support the notion that fast-twitch fibers may be as efficient, or perhaps more efficient than slow-twitch fibers during high power output exercises (33).

Economy of Movement and O_2 Cost of Transport

Parameters such as VO_{2max}, $\%VO_{2max}$, % efficiency, RER, and glucose and FFA flux rates are very important laboratory measures related to the energetics of human locomotion. However, our present capacities evolved without appreciation of those concepts or the ability to measure them. Further, while related to performance, they are nothing more than dependent variables measured during exercise. For this reason, some comparative physiologists (30, 125) and human physiologists (34) are particularly interested in understanding determinants of the "O_2 cost of transport" and the "economy of movement"; as measures of the metabolic costs of movement these parameters are defined as VO_2/distance and VO_2/speed.

For human runners, those with high levels of VO_{2max} may not be the most successful performers, especially if the VO_2 to run at a given speed is high, or in other words, economy is low (41). Seen in this context, a high VO_2 for an athlete running at a given speed represents a physiological strain, shifting energy substrate partitioning to greater reliance on CHO-derived fuels (Fig. 4). Because as of yet the existing evidence is that elite runners are neither extraordinarily gifted with regard to either cardiovascular capacity or muscle biochemistry (115), and because there is evidence of greater running economy in elite runners (76), at present it is reasonable to suspect that superior running economy has its basis in biomechanics.

In terms of the energetics of performance, it is clear that a greater economy of locomotion translates to a lesser cost of transport, with biomechanics as determinate. For instance, professional bicyclists can maintain speeds almost three times greater and for three times longer than Olympic marathon runners. As well, Olympic 100 m sprint runners are almost five times faster than are Olympic freestyle sprinters, but 500 m speed skaters are almost twice as fast as 400 m sprint runners. With assumptions of similar cardiovascular, metabolic, and muscle characteristics, in terms of performance, matters related to biomechanics become supremely important.

Summary and Conclusion

Human muscles, limbs and supporting ventilatory, cardiovascular, and metabolic systems are well adapted for walking, and there is reasonable transfer of efficiency of movement to bicycling. Our efficiency and economy of movement by walking ($\approx$ 30%) are far superior to those of apes. This

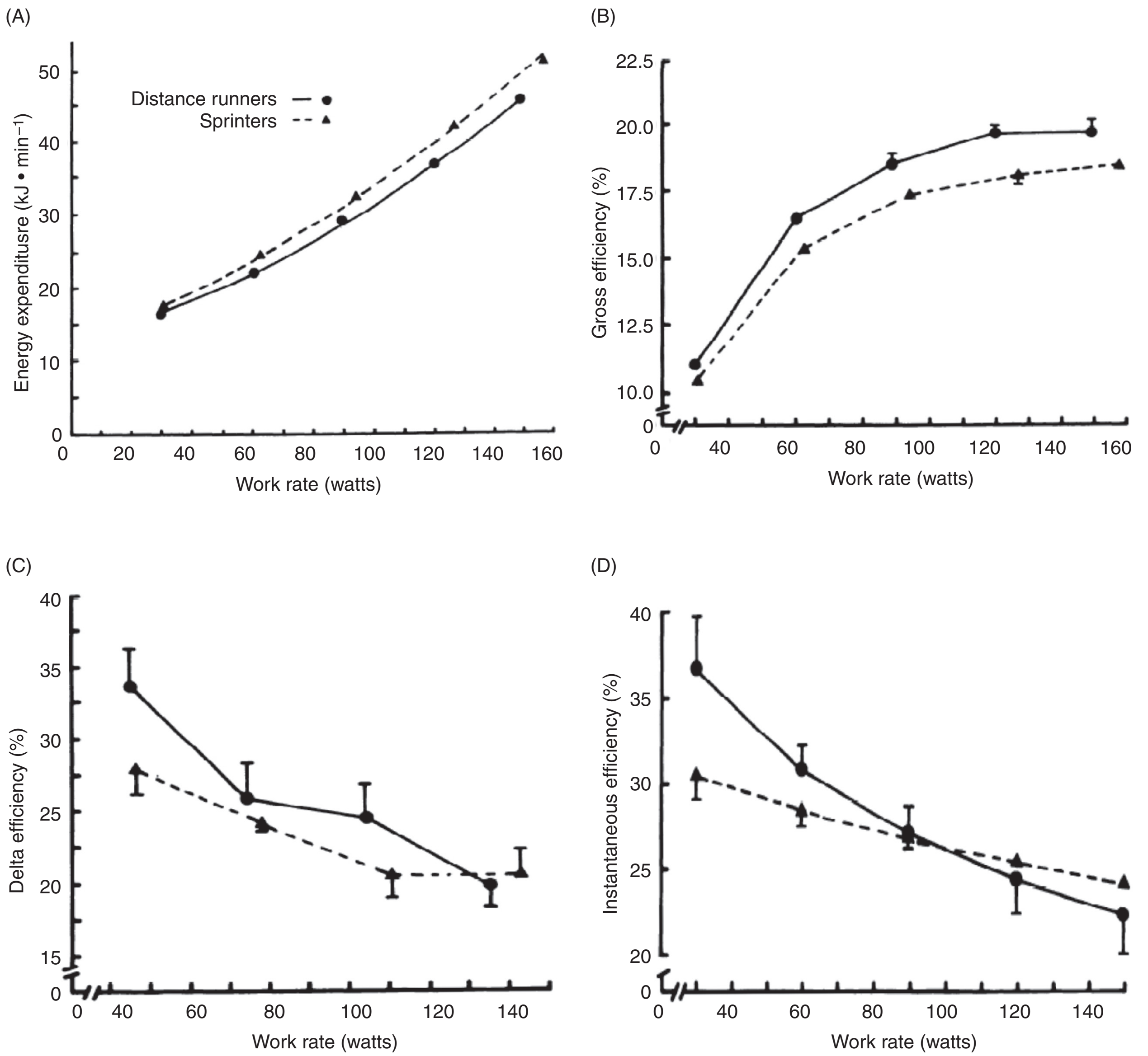

Figure 22 A compilation of results of Stuart et al. (123) who studied leg ergometer cycling efficiency of sprinters (fast twitch) and distance runners (slow twitch) athletes. (A) Higher rates of energy expenditure in sprinters exercising at given exercise power outputs indicate lesser efficiency. (B) Higher metabolic costs of exercise in sprinters makes for a lesser computed "gross" efficiency. (C, D) For both groups, delta (Δ) and instantaneous calculations show decreasing exercise efficiencies as exercise power outputs increase. Interestingly, while greater slopes of caloric expenditure regressed on power output make computed Δ and instantaneous efficiencies less at lower exercise power outputs, the computed efficiencies converge, or cross over at higher power outputs. Used with permission.

overall body efficiency during walking and bicycling represents the multiplicative interaction of a phosphorylative coupling efficiency of $\approx$ 60%, and a mechanical coupling efficiency of $\approx$ 50%. These coupling efficiencies compare well with those of other species adapted for locomotion. We are capable runners, but our speed and power are inferior to carnivorous and omnivorous terrestrial mammalian quadrupeds because of biomechanical and physiological constraints. But, because of our metabolic plasticity (i.e., the ability to switch among CHO- and lipid-derived energy sources), our endurance capacity is very good by comparison to most mammals, but inferior to highly adapted species such as wolves and migratory birds. Our ancestral ability for hunting and gathering depends on strategy and capabilities in the areas of thermoregulation, and metabolic plasticity. Clearly, our competitive advantage of survival in the biosphere depends in intelligence and behavior. Today, those abilities that served early hunter-gatherers make for interesting athletic competitions due to wide variations in human phenotypes. In contemporary society, the stresses of regular physical exercise serve to minimize morbidities and mortality associated with physical inactivity, overnutrition, and aging.

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References

1. Atwater WO, Benedict FG. *A Respiration Calorimeter with Appliances for the Direct Determination of Oxygen*. Washington, D.C.: Carnegie Institute of Washington, 1905.
2. Atwater WO, Rosa EB. *Description of a New Respiration Calorimeter and Experiments on the Conservation of Energy in the Human Body*, Bulletin 63 Washington, D.C.: U.S. Department of Agriculture, Office of Experiment Stations, Government Printing Office, 1899.
3. Awan MZ, Frearson N, Goldspink G, Waterson SE. Biochemical efficiency of smooth muscle and different types of striated muscle. *J Mechanochem Cell Motil* 1: 225-320, 1972.
4. Azevedo JL, Tietz E, Two-Feathers T, Paull J, Chapman K. Lactate, fructose and glucose oxidation profiles in sports drinks and the effect on exercise performance. *PLoS One* 2(9): e927, 2007.
5. Bangsbo J, Krustrup P, González-Alonso J, Saltin B. ATP production and efficiency of human skeletal muscle during intense exercise: Effect of previous exercise. *Am J Physiol Endocrinol Metab* 280: E956-E964, 2001.
6. Bárány M. ATPase activity of myosin correlated with speed of muscle shortening. *J Gen Physiol* 50: 197-218, 1967.
7. Bárány M, Close RI. The transformation of myosin in cross-innervated rat muscles. *J Physiol* 213: 455-474, 1971.
8. Barnard RJ, Edgerton VR, Furukawa T, Peter JB. Histochemical, biochemical, and contractile properties of red, white, and intermediate fibers. *Am J Physiol* 220: 410-414, 1971.
9. Benedict FG, Carpenter TM. *Respiration Calorimeters for Studying the Respiratory Exchange and Energy Transformations of Man*, Bulletin 123 Washington, D.C.: U.S. Department of Agriculture, Office of Experiment Stations, Government Printing Office, 1910.
10. Benedict FG, Smith HM. The Influence of athletic training upon basal metabolism. *Proc Natl Acad Sci U S A* 1: 102-103, 1915.
11. Bergman BC, Brooks GA. Respiratory gas exchange ratios during graded exercise in fed and fasted trained and untrained men. *J Appl Physiol* 86: 479-487, 1999.
12. Bergman BC, Butterfield GE, Wolfel EE, Casazza, Brooks GA. An evaluation of exercise and training on muscle lipid metabolism. *Am J Physiol* 276: E106-E117, 1999.
13. Bergman BC, Butterfield GE, Wolfel EE, Lopaschuk GD, Casazza GA, Horning MA, Brooks GA. Muscle net glucose uptake and glucose kinetics after endurance training in men. *Am J Physiol* 277: E81-E92, 1999.
14. Bergman BC, Horning MA, Casazza GA, Wolfel EE, Butterfield GE, Brooks GA. Endurance training increases gluconeogenesis during rest and exercise in men. *Am J Physiol Endocrinol Metab* 278: E244-251, 2000.
15. Bergman BC, Wolfel EE, Butterfield GE, Lopaschuk GD, Casazza GA, Horning MA, Brooks GA. Active muscle and whole body lactate kinetics after endurance training in men. *J Appl Physiol* 87: 1684-1696, 1999.
16. Bergström J. Local changes of ATP and phosphorylcreatine in human muscle tissue in connection with exercise. In: Chapman CB, editor. *Physiology of Muscular Exercise*. New York: American Heart Association, 1967, pp. 191-196. (Monograph No. 15).
17. Bramble DM, Lieberman DE. Endurance running and the evolution of Homo. *Nature* 432: 345-352, 2004.
18. Brand, M. The efficiency and plasticity of mitochondrial energy transduction. *Biochem Soc Trans* 33: 897-899, 2005.
19. Brooks GA. Master regulator or readout: The wisdom of distributed control. Focus on "Pyruvate suppresses PGC1alpha expression and substrate utilization despite increased respiratory chain content in C2C12 myotubes". *Am J Physiol Cell Physiol* 299: C216-C217, 2010.
20. Brooks GA, Butte NF, Rand WM, Caballero BH. Chronicle of the IOM physical activity recommendation: How a physical activity recommendation came to be among dietary recommendations. *Am J Clin Nutr* 79: 921S-930S, 2004.
21. Brooks GA, Fahey TD. *Exercise Physiology: Human Bioenergetics and its Applications*. New York: , John Wiley and Sons, 1984.
22. Brooks GA, Gladden LB. Metabolic systems: Non-oxidative (Glycolytic and phosphagen). In: Tipton CM, editor. *Exercise Physiology: People and Ideas*. American Physiological Society, 2003, pp. 322-360.
23. Brooks GA, Mercier J. The balance of carbohydrate and lipid utilization during exercise: The 'crossover' Concept, Brief Review. *J Appl Physiol* 76: 2253-2261, 1994.
24. Brooks GA, Trimmer JK. Literature supports the Crossover Concept. (Letter to the Editor). *J Appl Physiol* 80: 1073-1074, 1995.
25. Burke LM, Kiens B. "Fat adaptation" for athletic performance: The nail in the coffin? *J Appl Physiol* 100: 7-8, 2006.
26. Burke LM, Kiens B, Ivy JL. Carbohydrates and fat for training and recovery. *J Sports Sci* 22: 15-30, 2004.
27. Byrne NM, Hills AP, Hunter GR, Weinsier RL, Schutz Y. Metabolic equivalent: One size does not fit all. *J Appl Physiol* 99: 1112-1119.
28. Cathcart EP, Burnett WA. Influence of muscle work on metabolism of varying conditions of diet. *Proc Roy Soc (Biol)* 99: 405-426, 1926.
29. Casaburi R, Whipp BJ, Wasserman K, Koyal SN. Ventilatory and gas exchange responses to cycling with sinusoidally varying pedal rate. *J Appl Physiol* 44: 97-103, 1978.
30. Cavagna GA, Heglund NC, Taylor CR. Mechanical work in terrestrial locomotion: Two basic mechanisms for minimizing energy expenditure. *Am J Physiol* 233: R243-R261, 1977.
31. Cavagna GA, Kaneko M. Mechanical work and efficiency in level walking and running. *J Physiol* 268: 467-481, 1977.
32. Cherry-Girard, A. *The Worst Journey in the World: Antarctic 1910-1913*. Geo. H. Doran Publ., 1922.
33. Coetzer P, Noakes TD, Sanders B, Lambert MI, Bosch AN, Wiggins T, Dennis SC. Superior fatigue resistance of elite black South African distance runners. *J Appl Physiol* 75: 1822-1827, 1993.
34. Daniels J, Daniels N. Running economy of elite male and elite female runners. *Med Sci Sports Exerc* 24: 483-489, 1992.
35. Davies KJA, Packer L, Brooks GA. Biochemical adaptation of mitochondria, muscle, and whole animal respiration to endurance training. *Arch Biochem Biophys* 209: 539-559, 1981.
36. Donovan CM, Brooks GA. Muscular efficiency during steady-rate exercise II: Effects of walking speed and work rate. *J Appl Physiol* 43: 431-439, 1977.
37. Douglas CG. A method for determining the total respiratory exchange in man. *J Physiol* 42: 17-18, 1911.
38. Edwards RH, Dawson MJ, Wilkie DR, Gordon RE, Shaw D. Clinical use of nuclear magnetic resonance in the investigation of myopathy. *Lancet* 1(8274): 725-731, 1982.
39. Fan X, Hussien R, Brooks GA. H_2O_2-induced mitochondrial fission in C_2C_{12} myocytes. *Free Radicals in Biology and Medicine* 49: 1646-1654, 2010.
40. Faulkner JA. Physiology of swimming. *Res Q* 137: 41-54, 1966.
41. Faulkner JA, Daniels JT, Balke B. Effects of training at moderate altitude on physical performance capacity. *J Appl Physiol* 23: 85-89, 1967.
42. Fenn WO. A quantitative comparison between the energy liberated and the work performed by the isolated sartorius muscle of the frog. *J Physiol (London)* 58: 175-203, 1923.
43. Fiehn W, Peter JB. Properties of the fragmented sarcoplasmic reticulum from fast twitch and slow twitch muscles. *J Clin Invest* 50: 570-573, 1971.
44. Foster D. Banting lecture 1984. From glycogen to ketones-and back. *Diabetes* 33: 1188-1199, 1984.
45. Friedlander AL, Casazza GA, Horning MA, Brooks GA. Endurance training increases fatty acid turnover, but not fat oxidation, in young men. *J Appl Physiol* 86: 2097-2105, 1999.
46. Friedlander AL, Casazza GA, Horning MA, Budinger TF, Brooks GA. Effects of exercise intensity and training on lipid metabolism in young women. *Am J Physiol* 275: E853-E863, 1998.
47. Friedlander AL, Casazza GA, Horning MA, Huie MJ, Brooks GA. Training-induced alterations of glucose flux in men. *J Appl Physiol* 82: 1360-1369, 1997.
48. Friedlander AL, Casazza GA, Horning MA, Huie MJ, Piacentini MF, Trimmer JK, Brooks GA. Training-induced alterations of carbohydrate metabolism in young women: Women respond differently from men. *J Appl Physiol* 85: 1175-1186, 1998.
49. Friedlander AL, Jacobs KA, Fattor JA, Horning MA, Hagobian TA, Bauer TA, Wolfel EE, Brooks GA. Contributions of working muscle to whole body lipid metabolism vary with exercise intensity and training. *Am J Physiol Endocrinol Metab* 292: E107-E116, 2007.
50. Gaesser GA, Brooks GA. Muscular efficiency during steady-rate exercise: Effects of speed and work rate. *J Appl Physiol* 38: 1132-1139, 1975.
51. Gaesser GA, Brooks GA. Metabolic bases of excess post-exercise oxygen consumption: A review. *Med Sci Sports Exerc* 16: 29-43, 1984.
52. Gertz EW, Wisneski JA, Stanley WC, Neese RA. Myocardial substrate utilization during exercise in humans. Dual carbon-labeled carbohydrate isotope experiments. *J Clin Invest* 82: 2017-2025, 1988.
53. Gibbs CL, Gibson WR. Energy production of rat soleus muscle. *Am J Physiol* 223: 864-871, 1972.
54. Gladden LB. Lactate metabolism: A new paradigm for the third millennium. *J Physiol* 558: 5-30, 2004.

55. Gollnick PD, Armstrong RB, Saubert CW, Piehl K, Saltin B. Enzyme activity and fiber composition in skeletal muscle of untrained and trained men. *J Appl Physiol* 33: 312-319, 1972.
56. Griffin TM, Tolani NA, Kram R. Walking in simulated reduced gravity: Mechanical energy fluctuations and exchange. *J Appl Physiol* 86: 383-390, 1999
57. Haldane JS. A convenient form of gas analysis apparatus. *J Hyg (Lond)* 6: 74-76, 1906.
58. Hammel EP, Kronfeld DS, Ganjam VK, Dunlap HL Jr. Metabolic responses to exhaustive exercise in racing sled dogs fed diets containing medium, low, or zero carbohydrate. *Am J Clin Nutr* 30: 409-418, 1977.
59. Havemann L, West SJ, Goedecke JH, Macdonald IA, St Clair Gibson A, Noakes TD, Lambert EV. Fat adaptation followed by carbohydrate loading compromises high-intensity sprint performance. *J Appl Physiol* 100: 194-202, 2006.
60. Hawley JA, Burke LM. Carbohydrate availability and training adaptation: Effects on cell metabolism. *Exerc Sport Sci Rev* 38: 152-160, 2010.
61. Henderson SA, Black AL, Brooks GA. Effects of training on leucine turnover and oxidation during exercise. *Am J Physiol* 249: E137-E144, 1985.
62. Henderson GC, Fattor JA, Horning MA, Faghihnia N, Johnson ML, Luke-Zeitoun M, Brooks GA. Retention of intravenously infused [^{13}C]bicarbonate is transiently increased during recovery from exercise of hard but not moderate intensity. *J Appl Physiol* 103: 1604-1612, 2006.
63. Henderson GC, Fattor JA, Horning MA, Faghihnia N, Johnson ML, Luke-Zeitoun M, Brooks GA. More precise postexercise glucoregulation in women than men. *Am J Clin Nutr* 87: 1686-1694, 2008.
64. Henderson GC, Fattor JA, Horning MA, Faghihnia N, Johnson ML, Mau TL, Luke-Zeitoun M, Brooks GA. Lipolysis and free fatty acid metabolism during the post-exercise recovery period. *J Physiol* 584: 963-981, 2007.
65. Hill AV. The energy degraded in recovery processes of stimulated muscles. *J Physiol (Lond)* 46: 28-80, 1913.
66. Hill AV. Myothermic apparatus. *Proc R Soc Lon Ser B* 103: 117-137, 1928.
67. Hill AV, Long CNH, Lupton H. Muscular exercise, lactic acid and the supply and utilization of oxygen. Pt I-III. *Proc Royal Soc B* 96: 438-475, 1924a.
68. Hill AV, Long CNH, Lupton H. Muscular exercise, lactic acid and the supply and utilization of oxygen. Pt IV-VI. *Proc Royal Soc B* 97: 84-138, 1924b.
69. Hill AV, Long CNH, Lupton H. Muscular exercise, lactic acid and the supply and utilization of oxygen. Pt VII-VIII. *Proc Royal Soc B* 97: 155-176, 1924c.
70. Hill AV, Lupton H. Muscular exercise, lactic acid and the supply and utilization of oxygen. *Quar J Med* 16: 135-171, 1923.
71. Hinkle PC, Kumar MA, Resetar A, Harris DL. Mechanistic stoichiometry of mitochondrial oxidative phosphorylation. *Biochemistry* 30: 3576-3582, 1991
72. Hochachka PW. Metabolic biochemistry and the making of a mesopelagic mammal. *Experientia* 48: 570-575, 1992.
73. Hochachka PW. Oxygen, homeostasis, and metabolic regulation. *Adv Exp Med Biol* 475: 311-335, 2000.
74. Hochachka PW, Rupert JL, Monge C. Adaptation and conservation of physiological systems in the evolution of human hypoxia tolerance. *Comp Biochem Physiol A Mol Integr Physiol* 124: 1-17, 1999.
75. Holloszy JO. Biochemical adaptations in muscle. Effects of exercise on mitochondrial oxygen uptake and respiratory enzyme activity in skeletal muscle. *J Biol Chem* 242: 2278-2282, 1967.
76. Holmér I. Physiology of swimming man. *Exerc Sport Sci Rev* 7: 87-123, 1979.
77. Hultman E, Bergström J, Anderson NM. Breakdown and resynthesis of phosphorylcreatine and adenosine triphosphate in connection with muscular work in man. *Scand J Clin Lab Invest* 19: 56-66, 1967.
78. Hultman E, Spriet LL. Dietary intake prior to and during exercise. In: Horton ES, Terjung RL, editors. *Exercise, Nutrition and Energy Metabolism*. New York: Macmillan, 1988, p. 132-149.
79. Jacobs KA, Krauss RM, Fattor JA, Horning MA, Friedlander AL, Bauer TA, Hagobian TA, Wolfel EE, Brooks GA. Endurance training has little effect on active muscle fatty acid, lipoprotein, or triglyceride net balances. *Am J Physiol Endocrinol Metab* 29: E656-E665, 2006.
80. Jéquier E, Flatt J-P. Recent advances in human energetics. *NIPS* 1: 112-114, 1986.
81. Jeukendrup AE. Carbohydrate and exercise performance: The role of multiple transportable carbohydrates. *Curr Opin Clin Nutr Metab Care* 13: 452-457, 2010.
82. Johnson ML, Zarins Z, Fattor JA, Horning MA, Messonnier L, Lehman SL, Brooks GA. Twelve weeks of endurance training increases FFA mobilization and reesterification in postmenopausal women. *J Appl Physiol* 109: 1573-1581, 2010.
83. Jubrias SA, Vollestad NK, Gronka RK, Kushmerick MJ. Contraction coupling efficiency of human first dorsal interosseous muscle. *J Physiol* 586: 1993-2002, 2008.
84. Karlsson J. Lactate and phosphagen concentrations in working muscle in man. *Acta Physiol Scand* 358: 313-329, 1971.
85. Kirkwood SP, Munn EA, Packer L, Brooks GA. Mitochondrial reticulum in limb skeletal muscle. *Am J Physiol* 251: C395-C402, 1986.
86. Kleiber M. *The fire of life: An introduction to animal energetics*. New York: Wiley, 1961, pp. 116-128.
87. Krogh A, Lindhard J. The changes in respiration at the transition from work to rest. *J Physiol (Lond)* 53: 431-437, 1920.
88. Kuo CC, Fattor JA, Henderson GC, Brooks GA. Effect of exercise intensity on lipid oxidation in fit young adults during exercise recovery. *J Appl Physiol* 99: 349-356, 2005.
89. Kushmerick MJ, Conley KE. Energetics of muscle contraction: The whole is less than the sum of its parts. *Biochem Soc Trans* 30: 227-231, 2002.
90. Lee CP, Gu Q, Xiong Y, Mitchell RA, Ernster L. P/O ratios reassessed: Mitochondrial P/O ratios consistently exceed 1.5 with succinate and 2.5 with NAD-linked substrates. *FASEB J* 10: 345-350, 1996.
91. Lehninger AL. *Bioenergetics*. Menlo Park, CA: W.A. Benjamin, Inc., 1971.
92. Lloyd BB, Zacks RM. The mechanical efficiency of treadmill running against a horizontal impeding force. *J. Physiol (London)* 223: 355-363, 1972.
93. Lusk G. Animal calorimetry: Analysis of the oxidation of mixtures of carbohydrate and fat. *J Biol Chem* 59: 41-42, 1924.
94. Margaria R, Cerretelli P, Mangili F. Balance and kinetics of anaerobic energy release during strenuous exercise in man. *J Appl Physiol* 19: 623-628, 1964.
95. Margaria R, Edwards RHT, Dill DB. The possible mechanisms of contracting and paying the oxygen debt and the role of lactic acid in muscular contraction. *Am J Physiol* 106: 689-715, 1933.
96. Mazzeo RS, Brooks GA, Schoeller DA, Budinger TF. Disposal of [1-13C]-lactate during rest and exercise. *J Appl Physiol* 60: 232-241, 1986.
97. McKenzie S, Phillips SM, Carter SL, Lowther S, Gibala MJ, Tarnopolsky MA. Endurance exercise training attenuates leucine oxidation and BCOAD activation during exercise in humans. *Am J Physiol Endocrinol Metab* 278: E580-E587, 2000.
98. Mech LD, Boitani L. *Wolves: Behavior, ecology and conservation*. University of Chicago Press, 2003.
99. Millward DJ, Davies CT, Halliday D, Wolman SL, Matthews D, Rennie M. Effect of exercise on protein metabolism in humans as explored with stable isotopes. *Fed Proc* 41: 2686-2691, 1982.
100. Mommaerts WF. Energetics of muscular contraction. *Physiol Rev* 49: 427-508, 1969.
101. Nickleberry B, Brooks GA. No effect of cycling experience on leg cycle ergometer efficiency. *Med Sci Sports Exerc* 28: 1396-1401, 1996
102. Odessey R, Goldberg AL. Oxidation of leucine by rat skeletal muscle. *Am J Physiol* 223: 1376-1383, 1972.
103. Odland LM, Heigenhauser GJ, Wong D, Hollidge-Horvat MG, Spriet LL. Effects of increased fat availability on fat-carbohydrate interaction during prolonged exercise in men. *Am J Physiol* 274: R894-R902, 1998.
104. Onywera VO, Kiplamai FK, Boit MK, Pitsiladis YP. Food and macronutrient intake of elite Kenyan distance runners. *Int J Sport Nutr Exerc Metab* 14: 709-719, 2004.
105. Phinney SD, Bistrian BR, Evans WJ, Gervino E, Blackburn GL. The human metabolic response to chronic ketosis without caloric restriction: Preservation of submaximal exercise capability with reduced carbohydrate oxidation. *Metabolism* 32: 769-776, 1983.
106. Piersma T, Reneerkens J, Ramenofsky M. Baseline corticosterone peaks in shorebirds with maximal energy stores for migration: A general preparatory mechanism for rapid behavioral and metabolic transitions? *Gen Comp Endocrinol* 120: 118-126, 2000.
107. Poole DC, Gaesser GA, Hogan MC, Knight DR, Wagner PD. Pulmonary and leg VO_2 during submaximal exercise: Implications for muscular efficiency. *J Appl Physiol* 72: 805-810, 1992.
108. Pugh LG. The influence of wind resistance in running and walking and the mechanical efficiency of work against horizontal or vertical forces. *J Physiol (London)* 213: 255-276, 1971.
109. Ralston HJ. Energy-speed relation and optimal speed during level walking. *Intern. Z. Angew. Physiol* 17: 277-283, 1958.
110. Ralston HJ, Lukin L. Energy levels of human body segments during level walking. *Ergonomics* 12: 39-46, 1969.
111. Richardson RS, Noyszewski EA, Leigh JS, Wagner PD. Lactate efflux from exercising human skeletal muscle: Role of intracellular PO_2. *J Appl Physiol* 85: 627-634, 1998.
112. Richter EA, Kiens B, Saltin B, Christensen NJ, Savard G. Skeletal muscle glucose uptake during dynamic exercise in humans: Role of muscle mass. *Am. J. Physiol* 254E555-E561, 1988.
113. Roberts TJ, Weber JM, Hoppeler H, Weibel ER, Taylor CR. Design of the oxygen and substrate pathways. II. Defining the upper limits of carbohydrate and fat oxidation. *J Exp Biol* 199: 1651-1658, 1996.

114. Robergs RA, Ghiasvand F, Parker D. Biochemistry of exercise-induced metabolic acidosis. *Am J Physiol Regul Integr Comp Physiol* 287: R502-R516, 2004.
115. Saltin B, Kim CK, Terrados N, Larsen H, Svedenhag J, Rolf CJ. Morphology, enzyme activities and buffer capacity in leg muscles of Kenyan and Scandinavian runners. *Scand J Med Sci Sports* 5: 222-230, 1995.
116. Schmidt-Nielsen K. *Animal physiology: Adaptation and environment*. Cambridge, UK:. Cambridge University Press, 1997.
117. Schwatka F. *The long arctic search*. Mystic, CT: , Maine Historical Association, Inc., 1965.
118. Smith HM. *Gaseous exchange and physiological requirements for level walking*. Washington, D.C.: Carnegie Institution of Washington, 1922.
119. Spriet LL, Howlett RA, Heigenhauser GJF. An enzymatic approach to lactate production in human skeletal muscle during exercise. *Med Sci Sports Exerc* 32: 756-763, 2000.
120. Stainsby WN, Barclay JK. Exercise metabolism: O_2 deficit, steady level O_2 uptake and O_2 uptake for recovery. *Med Sci Sports* 2: 177-181, 1970.
121. Stanley WC, Gertz EW, Wisneski JA, Morris DL, Neese RA, Brooks GA. Systemic lactate turnover during graded exercise in man. *Am J Physiol* 249: E595-E602, 1985.
122. Stanley WC, Gertz EW, Wisneski JA, Neese RA, Morris DL, Brooks GA. Lactate metabolism in exercising human skeletal muscle: Evidence for lactate extraction during net lactate release. *J Appl Physiol* 60: 1116-1120, 1986.
123. Stuart MK, Howley ET, Gladden LB, Cox RH. Efficiency of trained subjects differing in maximal oxygen uptake and type of training. *J Appl Physiol* 50: 444-449, 1981.
124. Taylor CR, Rountree VJ. Running on two or on four legs: Which consumes more energy? *Science* 179: 186-187, 1973.
125. Taylor CR, Schmidt-Nielsen K, Raab JL. Scaling of energetic cost of running to body size in mammals. *Am J Physiol* 219, 1104-1107, 1970.
126. Todd KS, Butterfield GE, Calloway DH. Nitrogen balance in men with adequate and deficient energy intake at three levels of work. *J Nutr* 114: 2107-2118, 1984.
127. van Hall G, Strømstad M, Rasmussen P, Jans O, Zaar M, Gam C, Quistorff B, Secher NH, Nielsen HB. Blood lactate is an important energy source for the human brain. *J Cereb Blood Flow Metab* 29: 1121-1129, 2008.
128. Venables MC, Achten J, Jeukendrup AE. Determinants of fat oxidation during exercise in healthy men and women: A cross-sectional study. *J Appl Physiol* 98: 160-167, 2005.
129. Wallis GA, Friedlander AL, Jacobs KA, Horning MA, Fattor JA, Wolfel EE, Lopaschuk GD, Brooks GA. Substantial working muscle glycerol turnover during two-legged cycle ergometry. *Am J Physiol Endocrinol Metab* 293: E950-E957, 2007.
130. Watt MJ, Heigenhauser GJ, Spriet LL. Intramuscular triacylglycerol utilization in human skeletal muscle during exercise: Is there a controversy? *J Appl Physiol* 93: 1185-1195, 2002.
131. Wendt IR, Gibbs CL. Energy production of rat extensor digitorum longus muscle. *Am J Physiol* 224: 1081-1086, 1973.
132. Whipp BJ, Wasserman K. Efficiency of muscular work. *J Appl Physiol* 26: 644-648, 1969.
133. White TP, Brooks GA. [U-14C]-glucose, -alanine, and -leucine oxidation in rats at rest and two intensities of running. *Am J Physiol* 240: E155-E165, 1981.
134. Willis WT, Jackman MR. Mitochondrial function during heavy exercise. *Med Sci Sports Exerc* 26: 1347-1353, 1994.
135. Wolfe RR, Wolfe MH, Nadel ER, Shaw JH. Isotopic determination of amino acid-urea interactions in exercise in humans. *J Appl Physiol* 56: 221-229, 1984.
136. Wolfel EE, Groves BM, Brooks GA, Butterfield GE, Mazzeo RS, Moore LG, Sutton JR, Bender PR, Dahms TE, McCullough RE, Reeves JT et al. Oxygen transport during steady-state submaximal exercise in chronic hypoxia. *J Appl Physiol* 70: 1129-1136, 1991.
137. Zuntz N, Schumburg D. *Studien Zu Einer Physiologie Des Marches*. Berlin: Verlag von August Hirchwald, 1901.